Electromagnetic Calorimetry for the Gifted sPHENIXian

Anthony Hodges

sPHENIX Tea Talks

Tuesday, February 10th, 2026

NSF Ascend Fellow

Who Am I?

- Anthony Hodges, Postdoctoral Research Fellow, UIUC/NSF
- Formerly based on LI, now based in Brooklyn
  - Which is still technically on LI but don’t tell NYC people that
- Calorimeter Calibrations co-convener for 2 years
  - Creator of some of the first (poorly) calibrated calorimeter DST’s
- Heavy on the expert list for Run 25, less so Run 26
- Chair of sPHENIX PPG01, co-chair of PPG09 - all calorimeter measurements

The Goal of This Talk

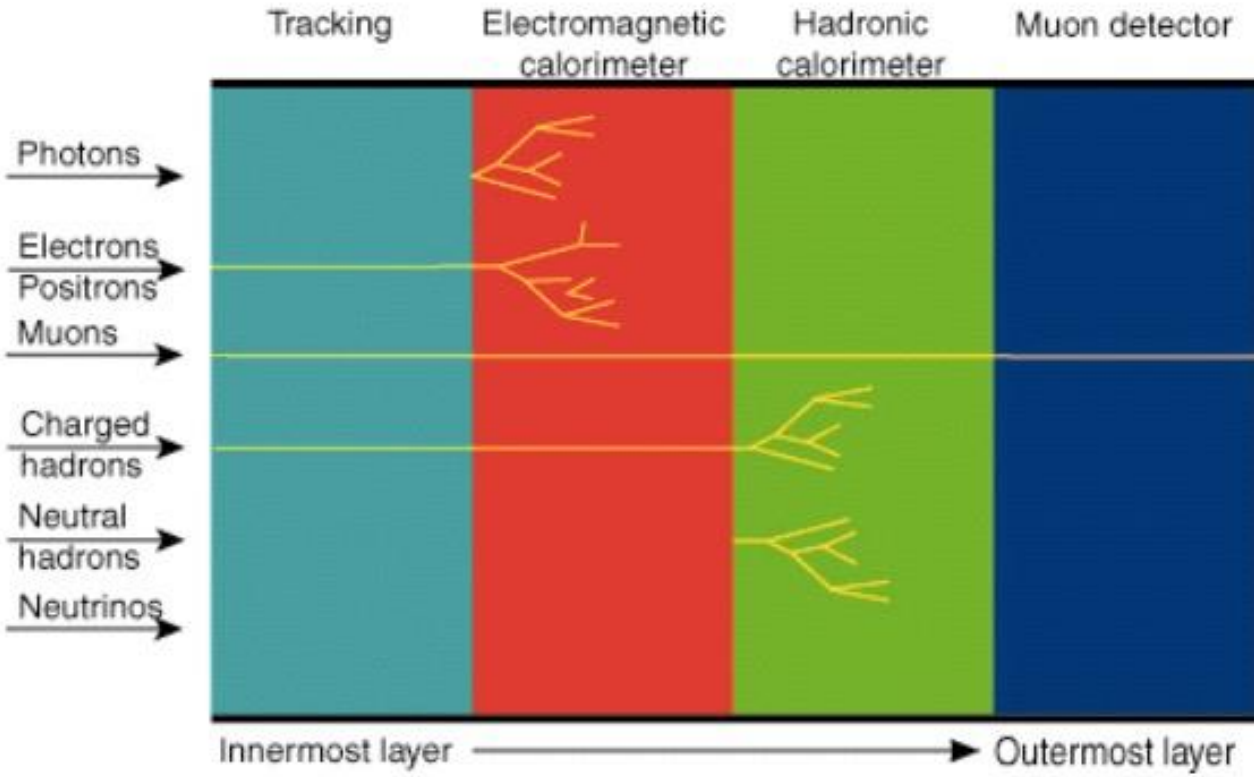
- **Not** to make you calorimeter experts
  - Too little time, too much information
- **Not** to convince tracking folks to abandon the tracking effort and join the calorimeter workforce
- **To** convince you that calorimetry is pretty freaking rad and that you need more of it in your life
- **To** give you a baseline amount of knowledge that will make calorimetric talks a little more grounded

What Are We Going To Talk About

- Calorimetry writ large
- The sPHENIX EMCal
- Calibrations
- Peculiarities you will see analyzing EMCal data
  - Mini-tutorial

What is Calorimetry?

- Broadly, measuring a particle's **energy**
- Large, dense materials **absorb** the energy of incoming particles
  - Calorimetry is a "destructive" measurement
- Readout converts this energy into a **signal**
- Calibration converts signal into a **measurement**



https://www.physicsmasterclasses.org/exercises/keyhole/en/main/JumpToExercises.html

Why is Calorimetry?

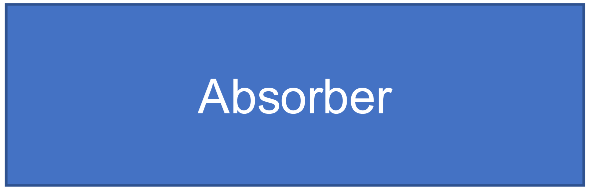
- Neutral particles: γ, π^0, η
  - "Unbiased" measurements of many phenomena
- Energy resolution: increases with increasing p_T → kinda stuff we want to measure (jets, high p_T photons)
- Building/budget constraints (cost/volume)
- Structural
  - sPHENIX *is* the outer HCal
  - Inner HCal's most important job is actually holding the EMCal
- Provides rare-probe triggers (jets, photons)

Let’s Make a Calorimeter

What could go wrong

Calorimetric Design

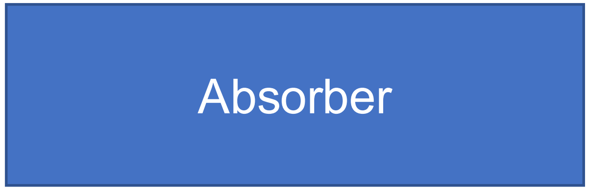
- The calorimeter's job is to eat energy
- Basic formula:
  - Absorber: high density material to induce EM showering



Calorimetric Design

- The calorimeter's job is to eat energy
- Basic formula:
    - Absorber: high density material to induce EM showering

Good absorber materials:
Lead, Steel, Tungsten



Calorimetric Design

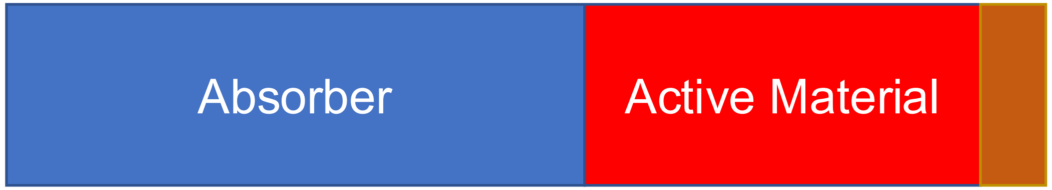
- The calorimeter’s job is to eat energy
- Basic formula:
  - Absorber: high density material to induce EM showering
  - Active material: Responds to energy embedded in it, creates signal for readout

Polystyrene + Scintillator dopant

Calorimetric Design

- The calorimeter's job is to eat energy
- Basic formula:
  - Absorber: high density material to induce EM showering
  - Active material: Responds to energy embedded in it, creates signal for readout
  - Readout: You guessed it, reads out signal from active material

SiPM's, PMT's, etc.



Readout

Calorimetric Design

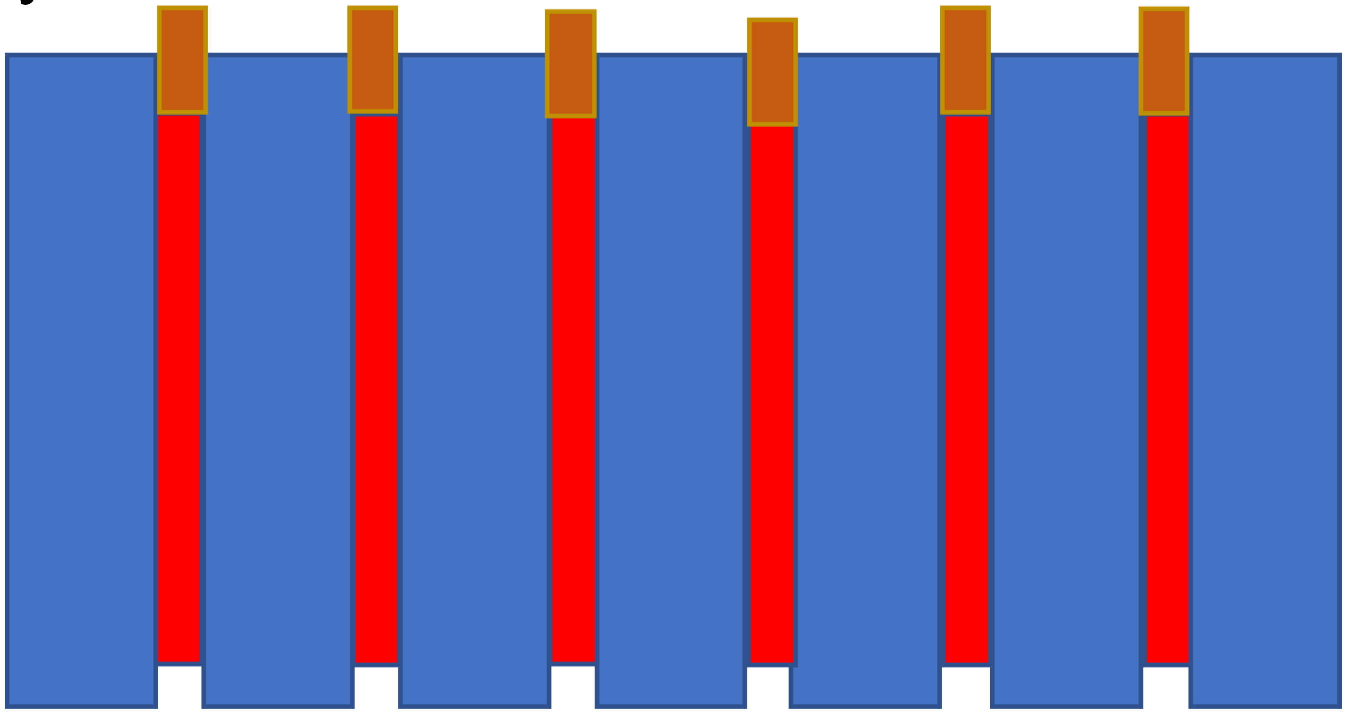
- The calorimeter’s job is to eat energy
- Basic formula:
  - Absorber: high density material to induce EM showering
  - Active material: Responds to energy embedded in it, creates signal for readout
  - Readout: You guessed it, reads out signal from active material

SiPM’s, PMT’s, etc.

Right pieces, Bad Design!

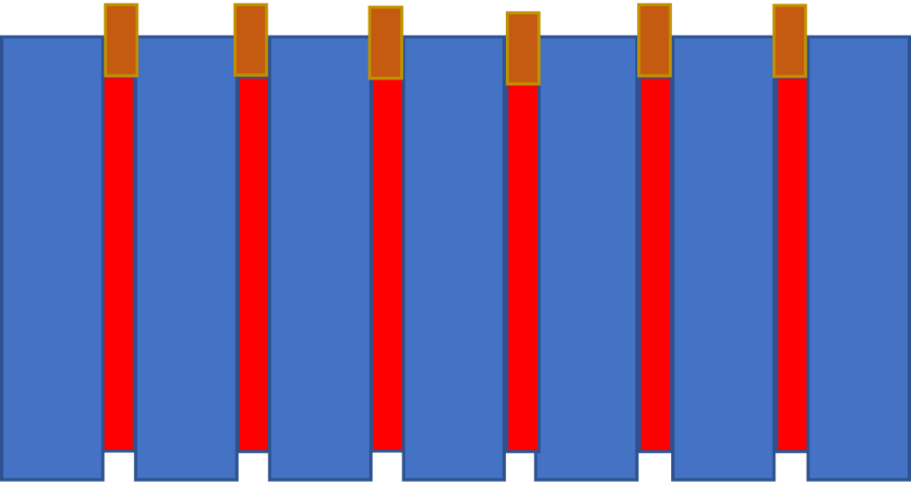
Calorimetric Design

- Intersperse **absorber** with **active material** so you can **sample** different stages of the shower → you've created a sampling calorimeter!
- Compared to a *homogeneous* calorimeter
- Give each layers of active material its own readout

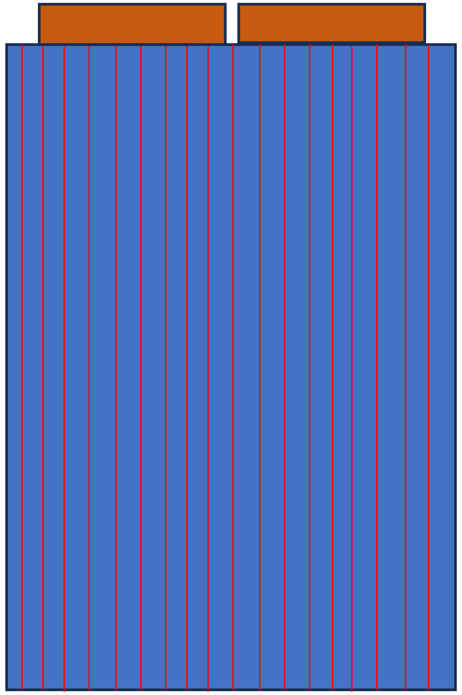


Calorimetric Design

- Intersperse **absorber** with **active material** so you can **sample** different stages of the shower
- Give each layers of active material its own readout



Note: This is basically the HCal



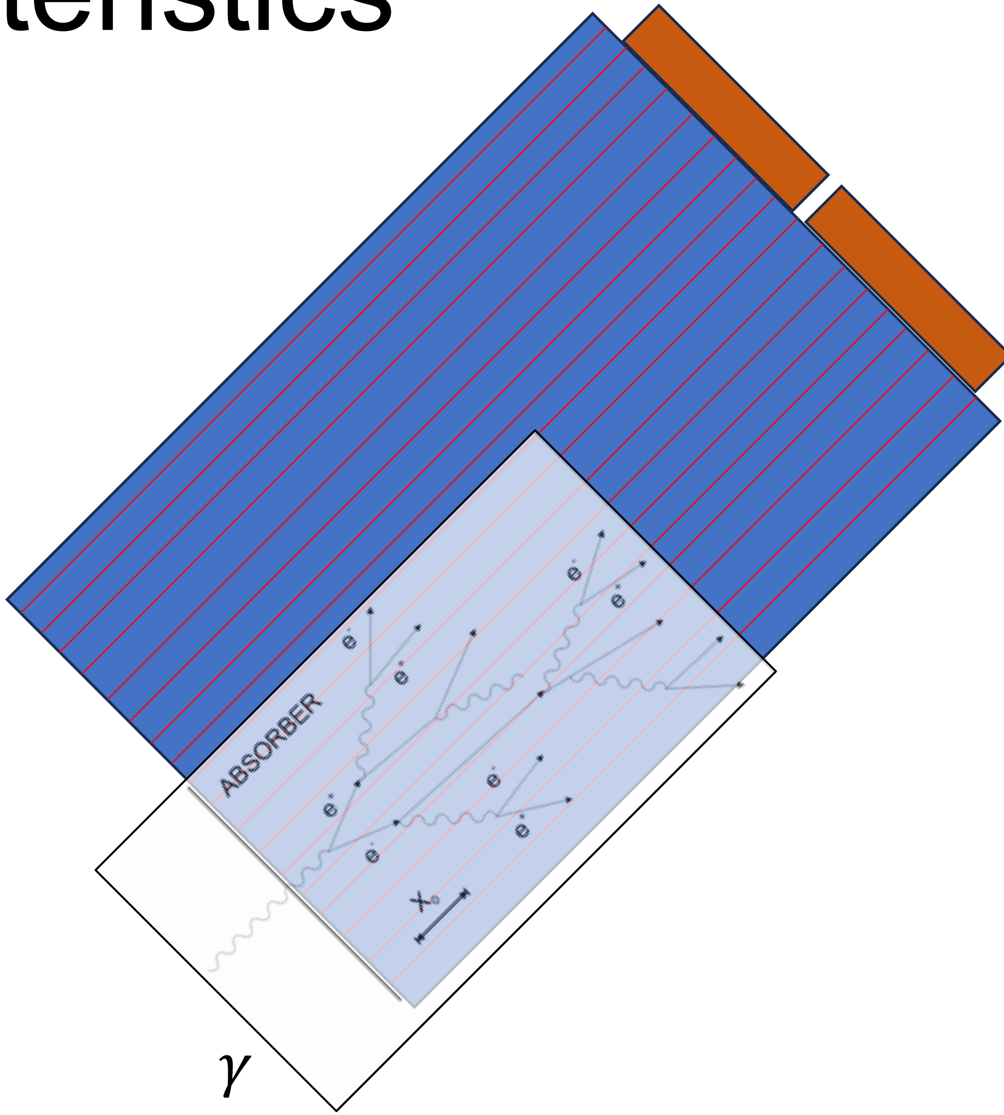
This is more like the EMCal

Sampling Calorimetry Characteristics

- d: absorber depth
- s: active material depth
- Gives rise to the “sampling fraction”:

$$f_{sample} = \frac{d \cdot dE/dx_{MIP}^{active}}{d \cdot dE/dx_{MIP}^{Absorber} + s \cdot dE/dx_{MIP}^{Active}}$$

- Typical sampling fraction ~few %

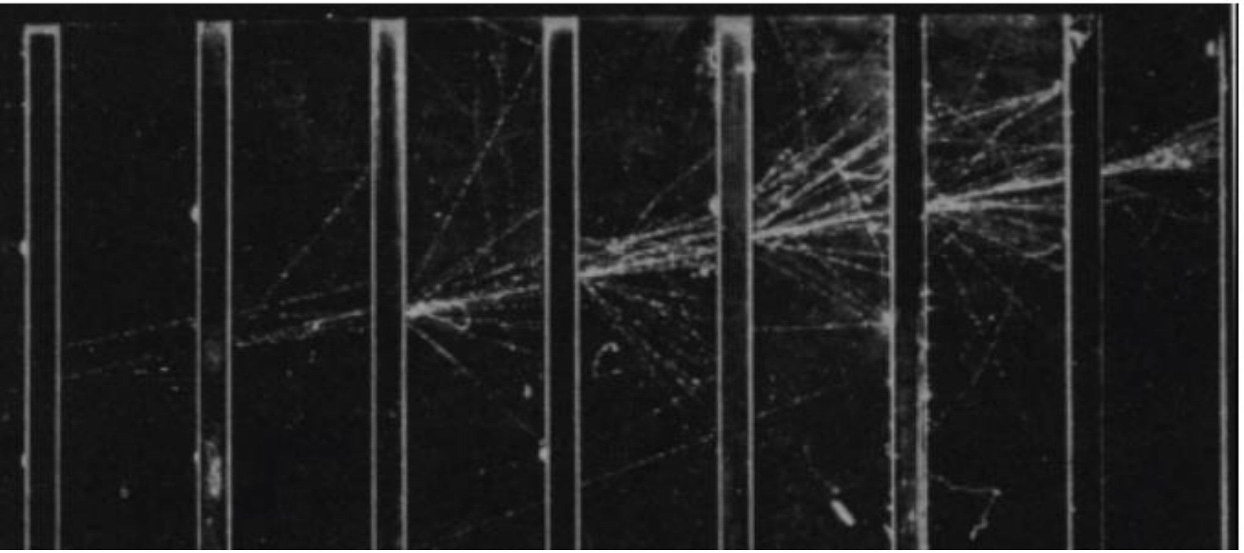


Electromagnetic Particle Showers

Why do they need to shower if they don't have noses?

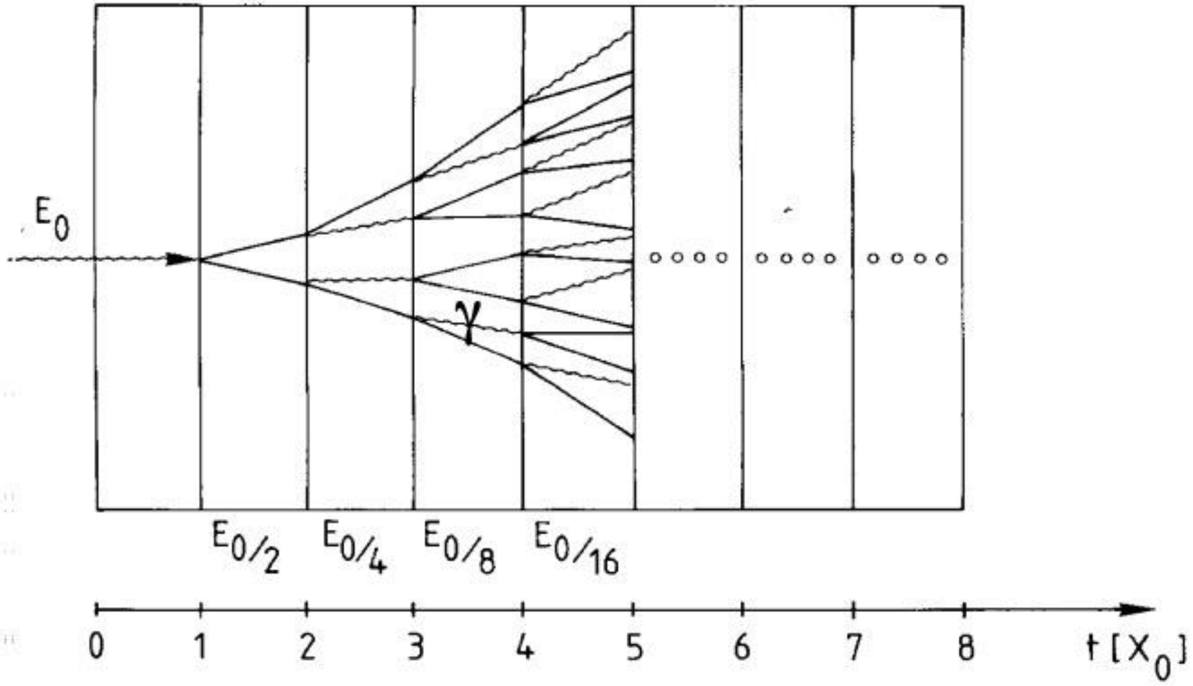
Electromagnetic Showers

Electron shower in cloud chamber
using lead absorbers



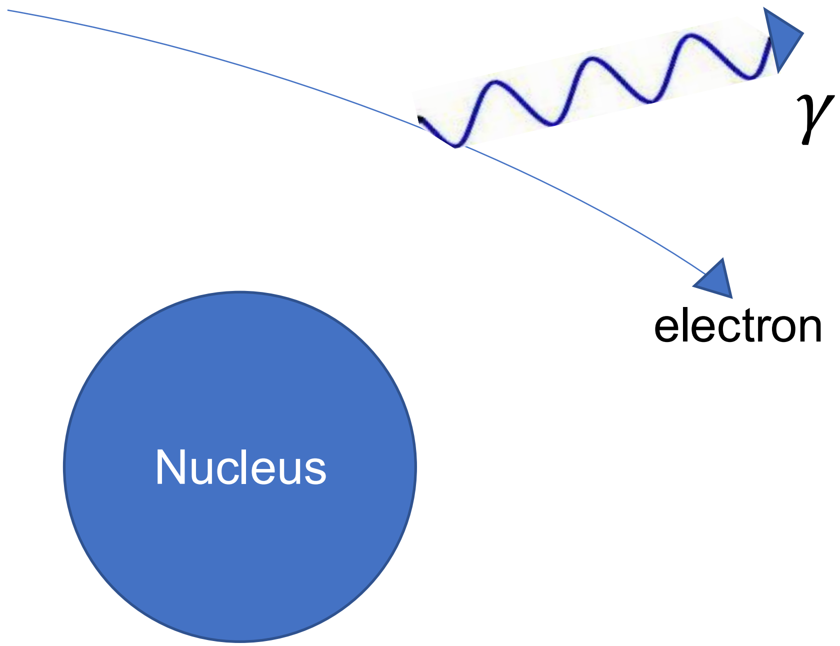
Christian Joram, [CERN Summer Student Lectures]

Basic model of an EM shower



Electromagnetic Showers

- For **electrons**
- Bremsstrahlung, “breaking radiation”



$$-\frac{dE}{dx} = \underbrace{E4\alpha N_A \frac{Z^2}{A} r_e^2 \ln\left(\frac{183}{Z^{\frac{1}{3}}}\right)}$$

$$X_0 = \frac{A}{4\alpha N_A Z^2 r_e^2 \ln\left(\frac{183}{Z^{\frac{1}{3}}}\right)}$$

“Radiation Length”

Electromagnetic Showers

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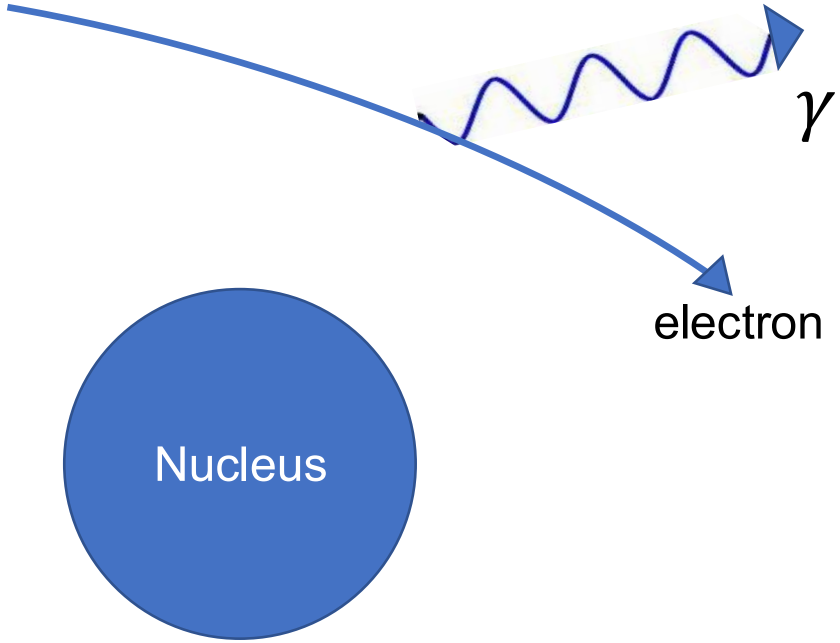

$$X_0 = \frac{A}{4\alpha N_A Z^2 r_e^2 \ln\left(\frac{183}{Z^{\frac{1}{3}}}\right)} \rightarrow -\frac{dE}{dx} = \frac{E}{X_0}$$

“Radiation Length” - Average distance to lose 63.2% of energy (all but $1/e$)

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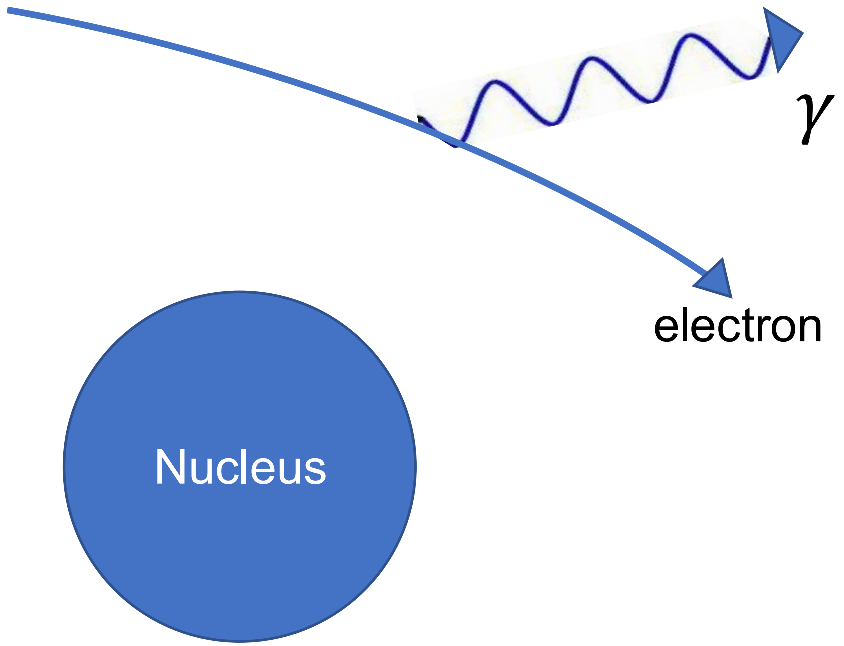

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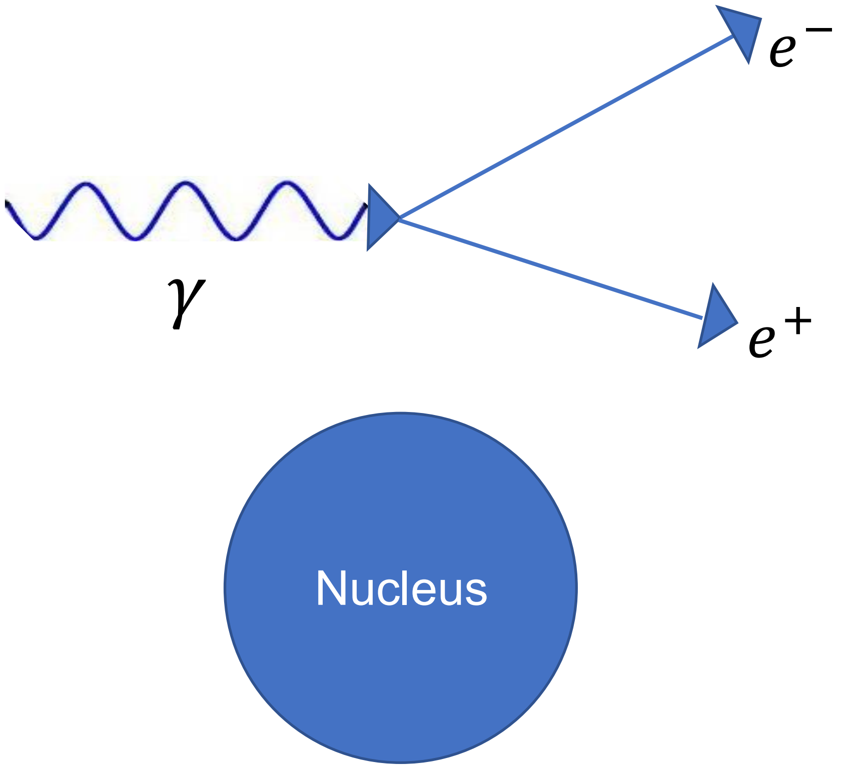
Denser materials make for shorter radiation lengths

Electromagnetic Showers

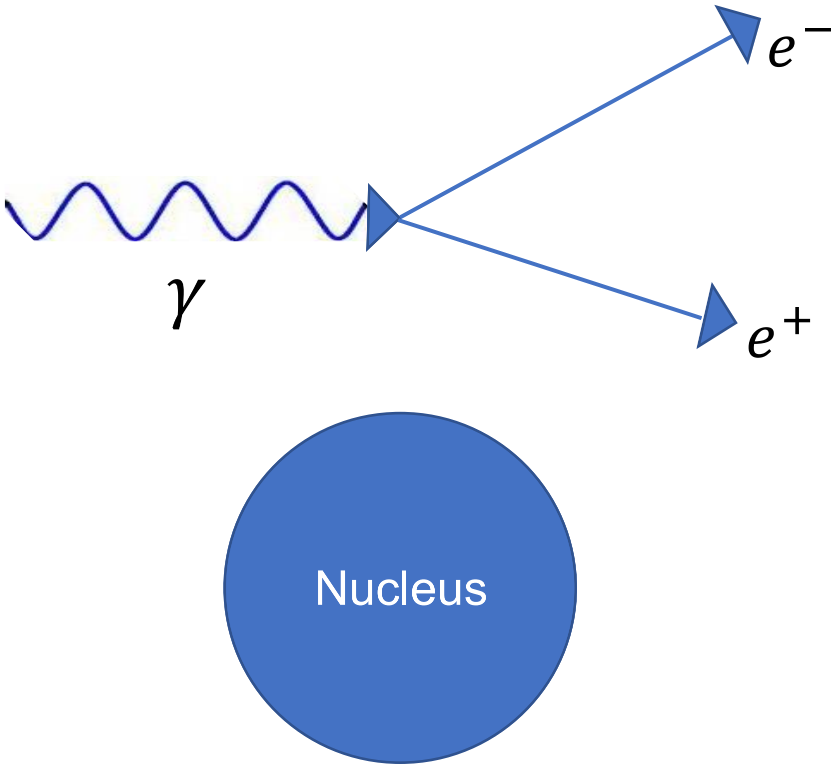
- For **photons**
- Pair creation, $\gamma \rightarrow e^{+} + e^{-}$



Electromagnetic Showers

- For **photons**
- Pair creation, $\gamma \rightarrow e^+ + e^-$

$$\sigma_{pair} \approx \frac{7}{9} 4\alpha Z^2 r_e^2 \ln\left(\frac{183}{Z^{\frac{1}{3}}}\right)$$



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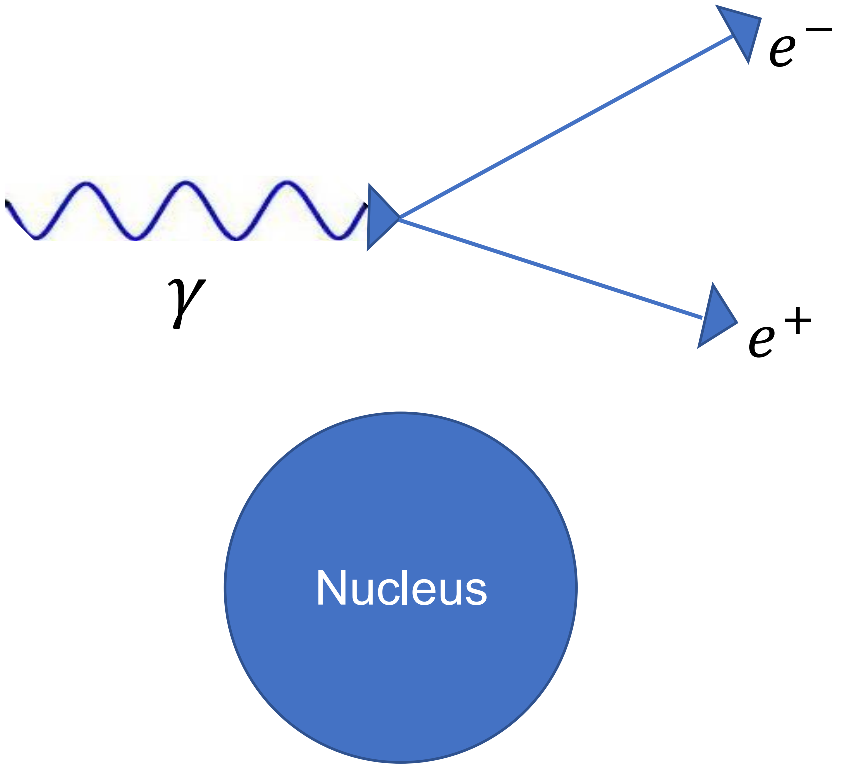

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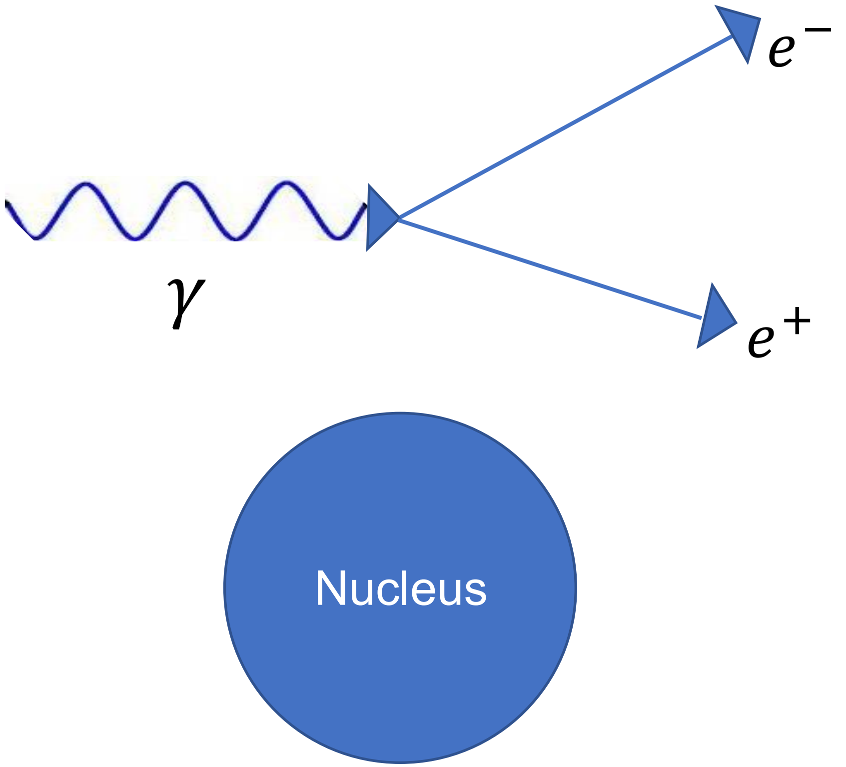

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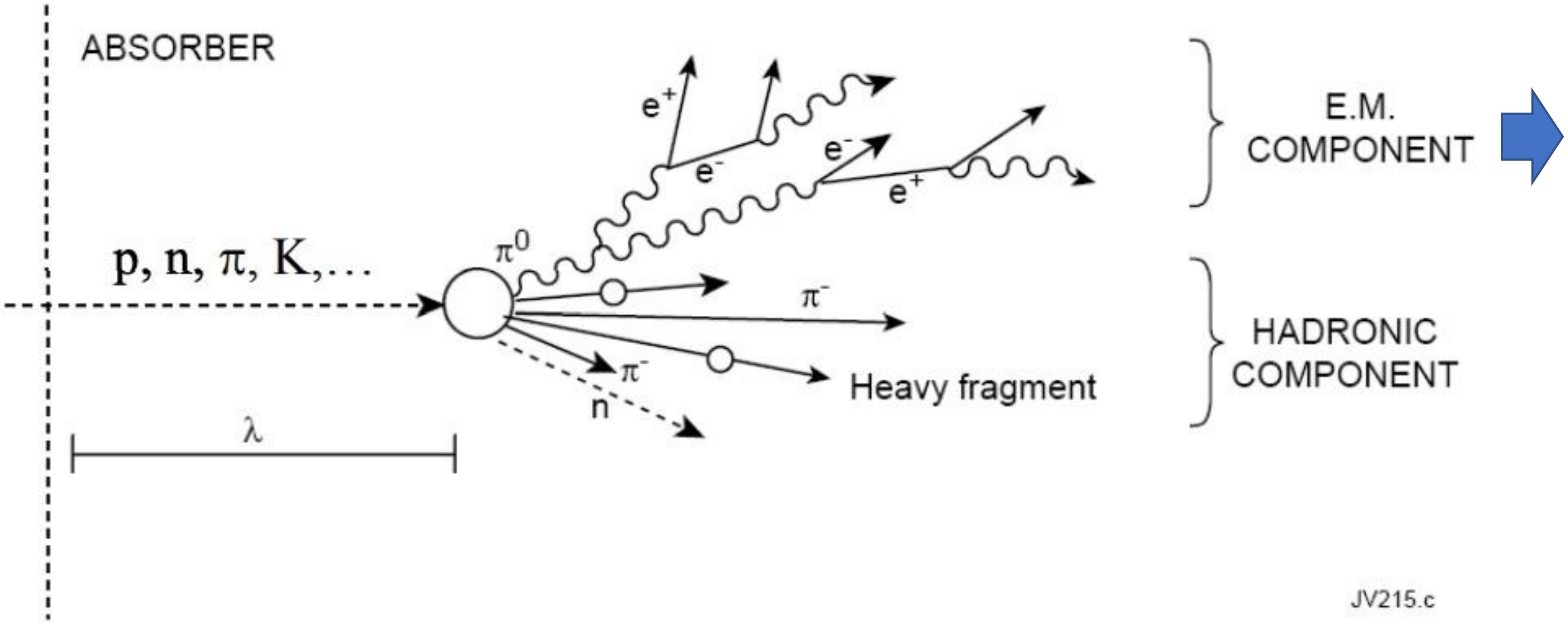
Our good friend, the radiation length

Hadronic Showers

I already made a joke about this

Hadronic Showers

- Nasty business, much more complicated!



From π^0 decay
Familiar particles and shower development

David Cockerill, [Introduction to Calorimeters]

Hadronic Showers

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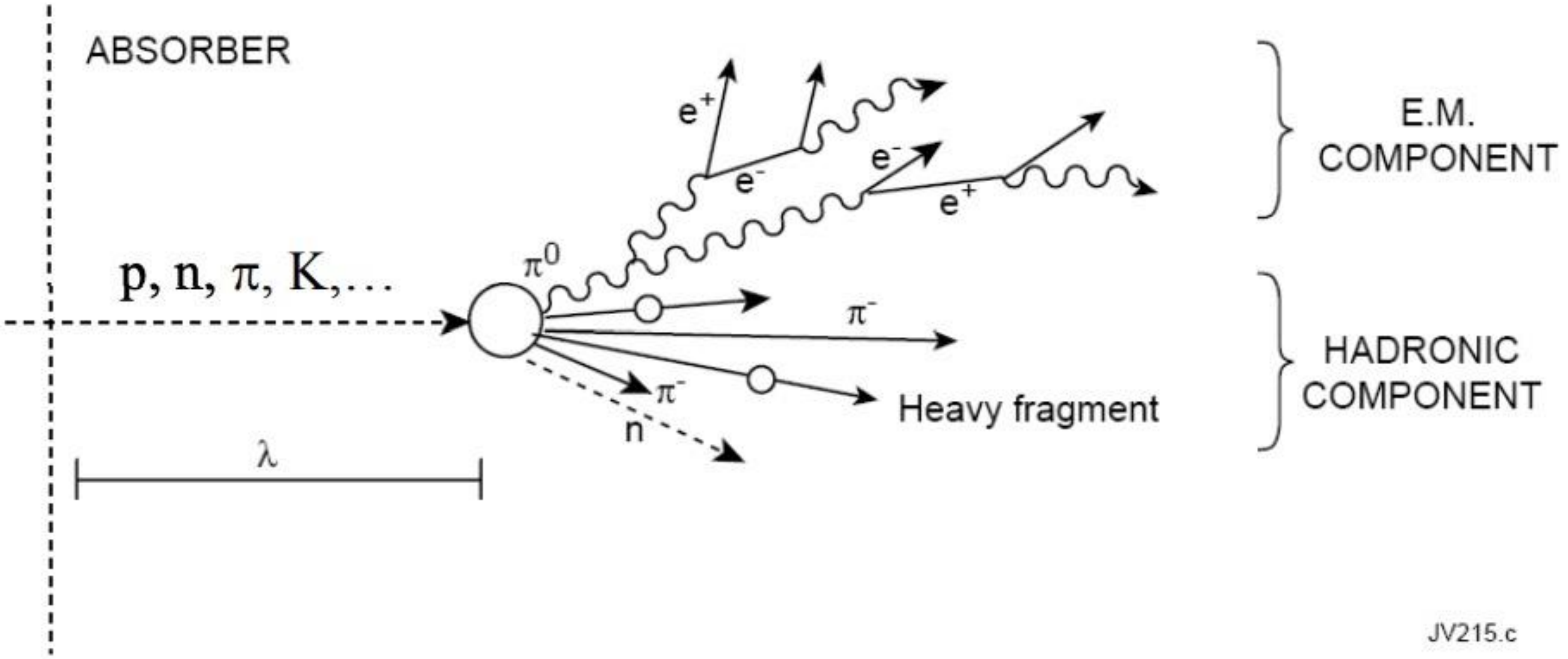
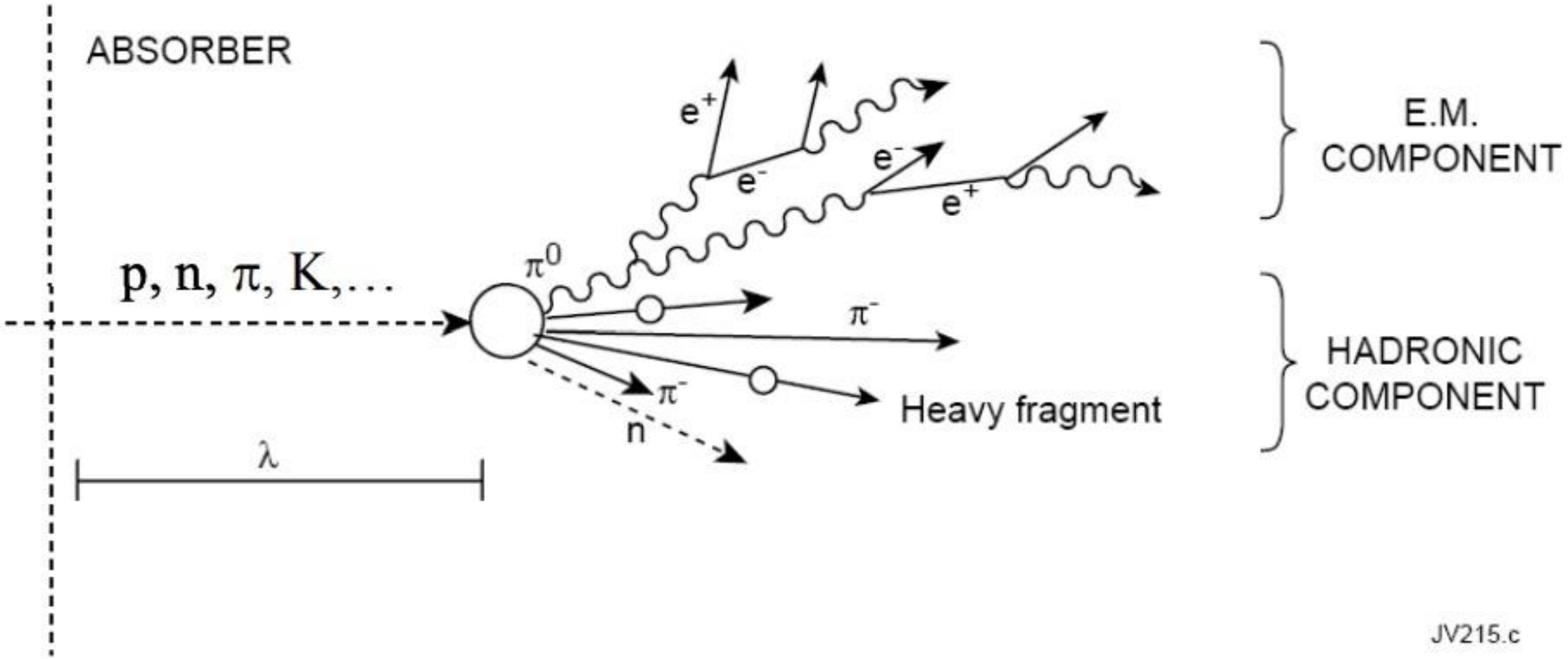


Nuclear interactions.
Not measured

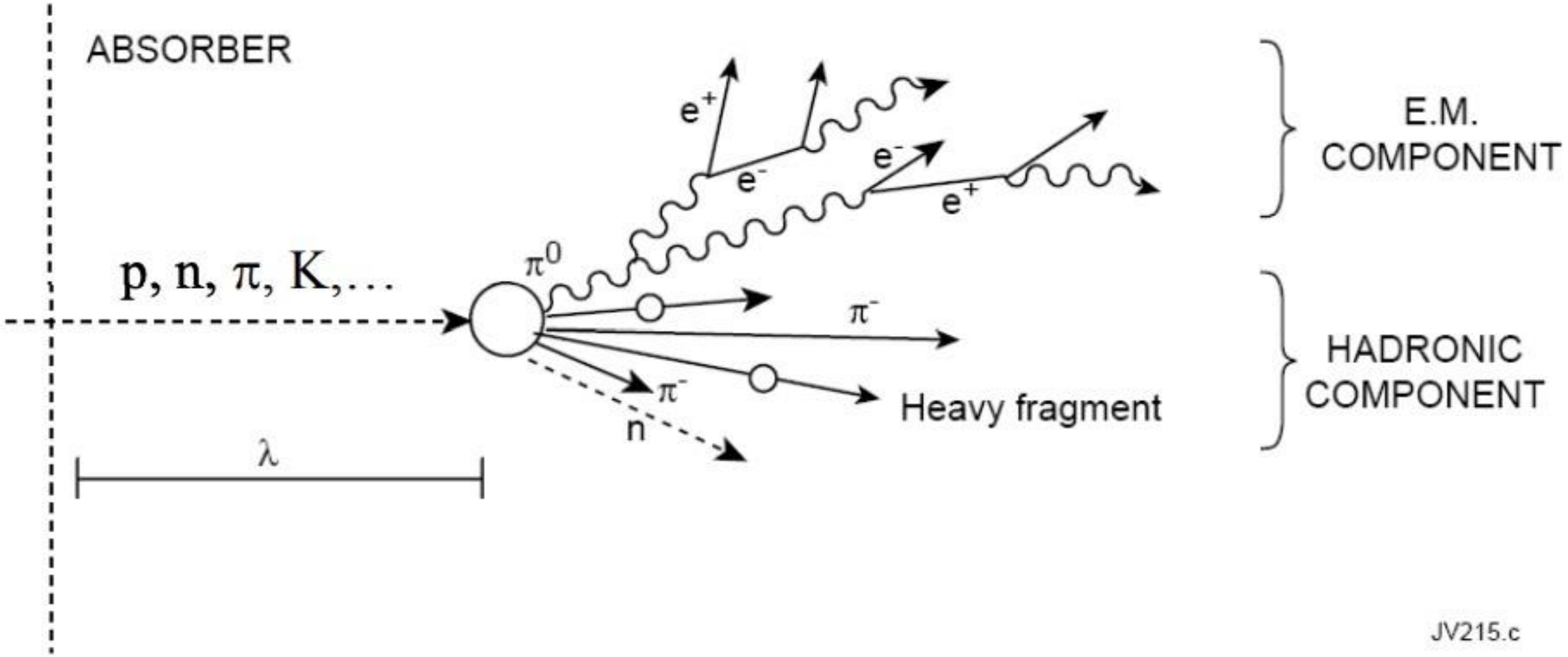
David Cockerill, [Introduction to Calorimeters]

Hadronic Showers



Principle characteristic: **Nuclear Interaction Length,** $\boldsymbol{\lambda_l} \propto 35A^{\frac{1}{3}}$

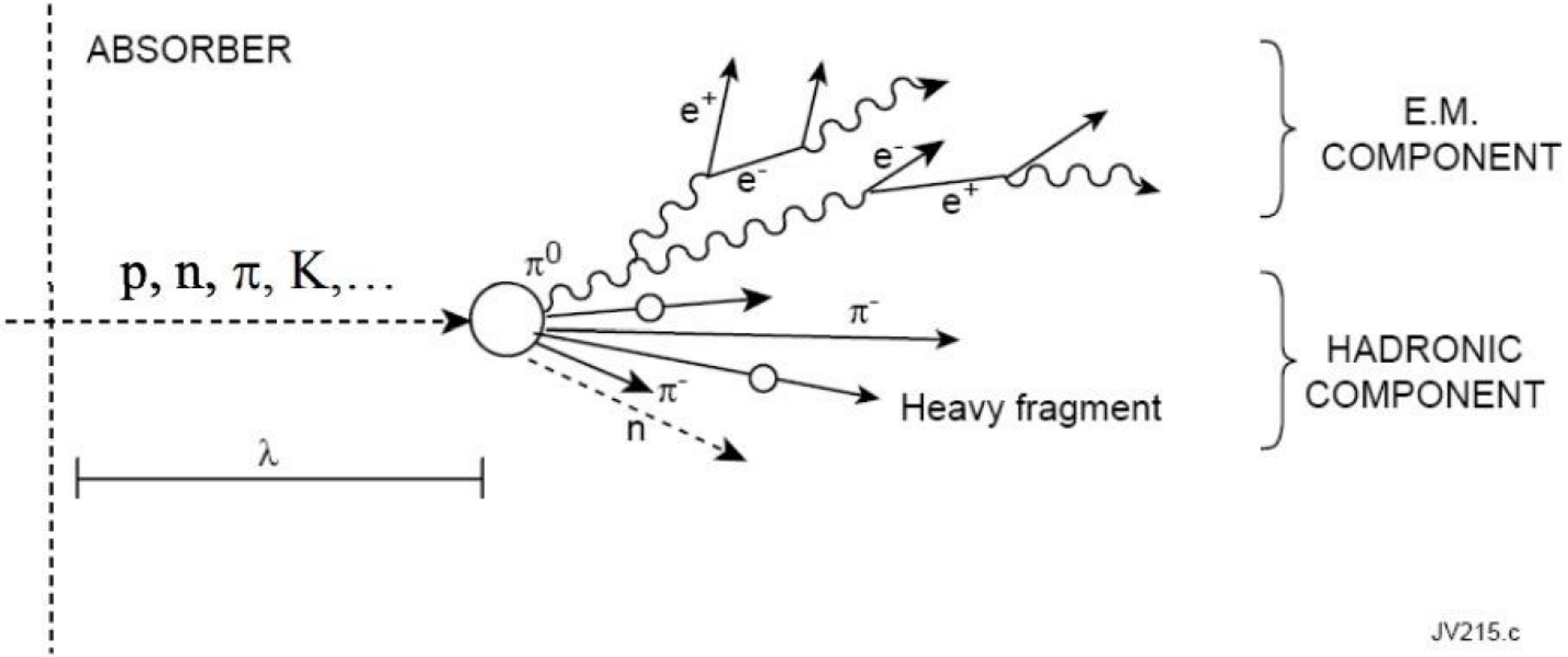
Hadronic Showers



Principle characteristic: **Nuclear Interaction Length,** $\lambda_l \propto 35A^{\frac{1}{3}}$

"Average distance travelled before undergoing inelastic collision"

Hadronic Showers



Principle characteristic: **Nuclear Interaction Length,** $\boldsymbol{\lambda_l} \propto 35A^{\frac{1}{3}}(1/\rho)$

Iron ≈17cm
Steel ≈ 14cm
Tungsten ≈ 10cm

Hadronic and Electromagnetic Responses

- In general, the response of **calorimeters** is different whether you have a **hadron** or something like an **electron**
- We refer to this as the **e/h** ratio
- e/h is usually > 1, as a calorimeter is going to sample more of the energy from an electromagnetic shower than a hadronic one
- **Compensating calorimeters** attempt to get e/h ≈ 1
  - Generally by carefully adjusting the interplay between absorbing and active materials
- The e/h ratio for a calorimeter typically plays a **befuddling** role in calibrations
  - I will not elaborate on this

What It All Means

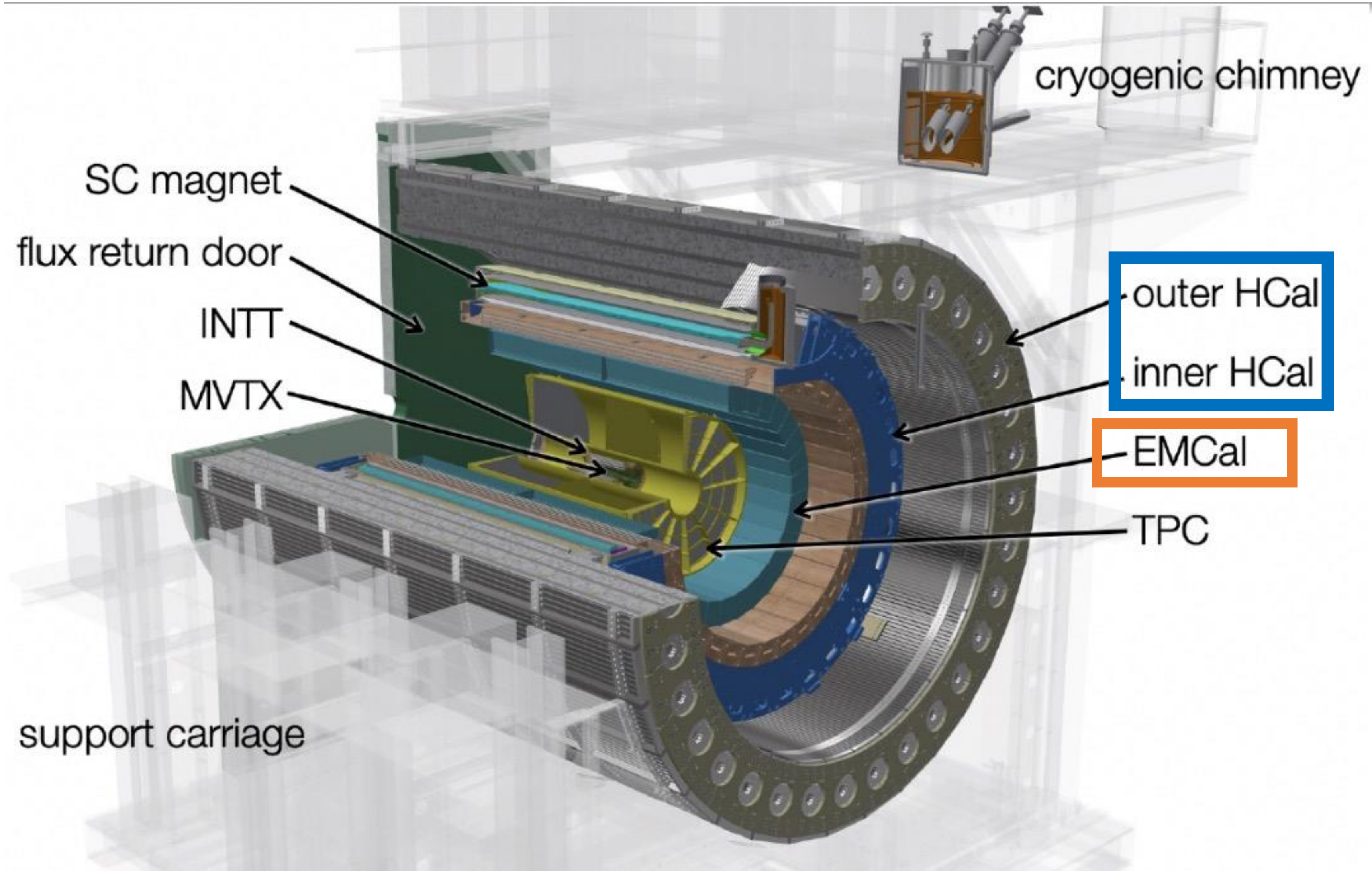
- You want super dense materials to build calorimeters out of because it means
  1. There's a higher chance of shower initiation within the material
  2. There's a greater amount of energy eaten by the calorimeter per generation of the shower
- Why don't we just build big honkin chunks of steel or tungsten or whatever? Have infinite numbers of radiation lengths?
  - Money, space, structural engineering.

The sPHENIX EMCal

The GOAT

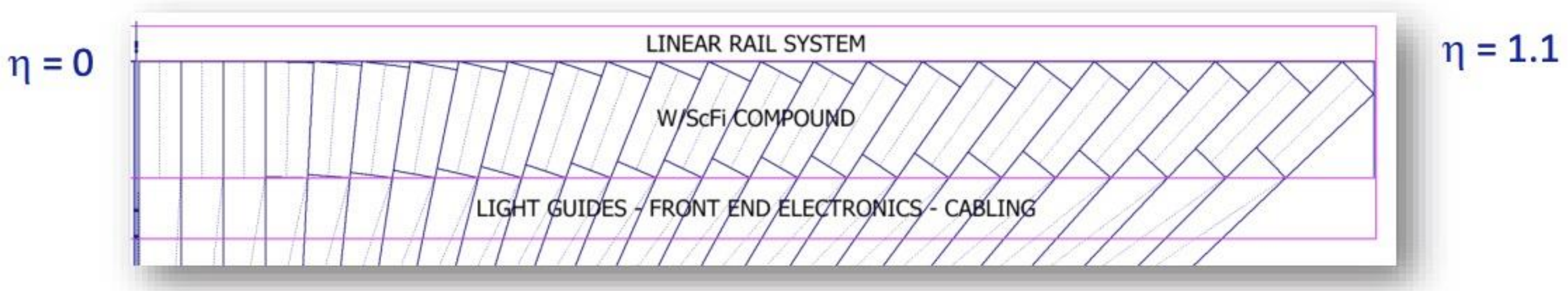
Calorimetry - sPHENIX

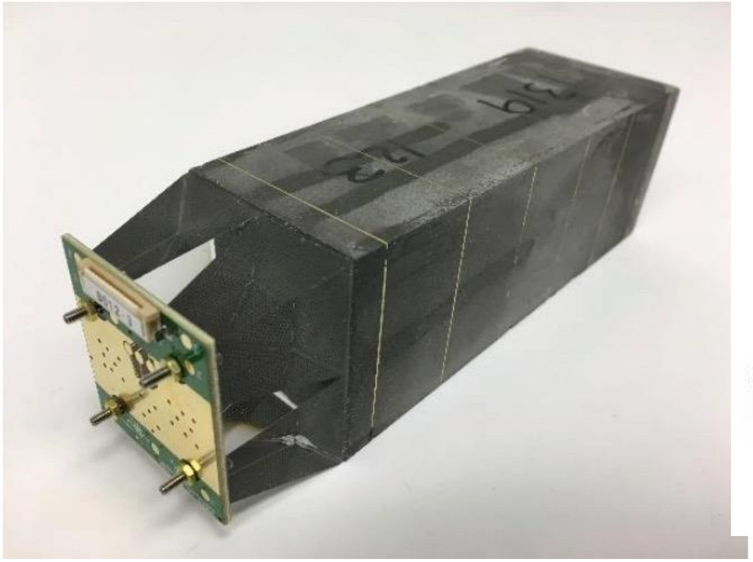
- Combines both electromagnetic **and** hadronic calorimetry
- HCal **segmented** into inner and outer calorimeters separated by magnet

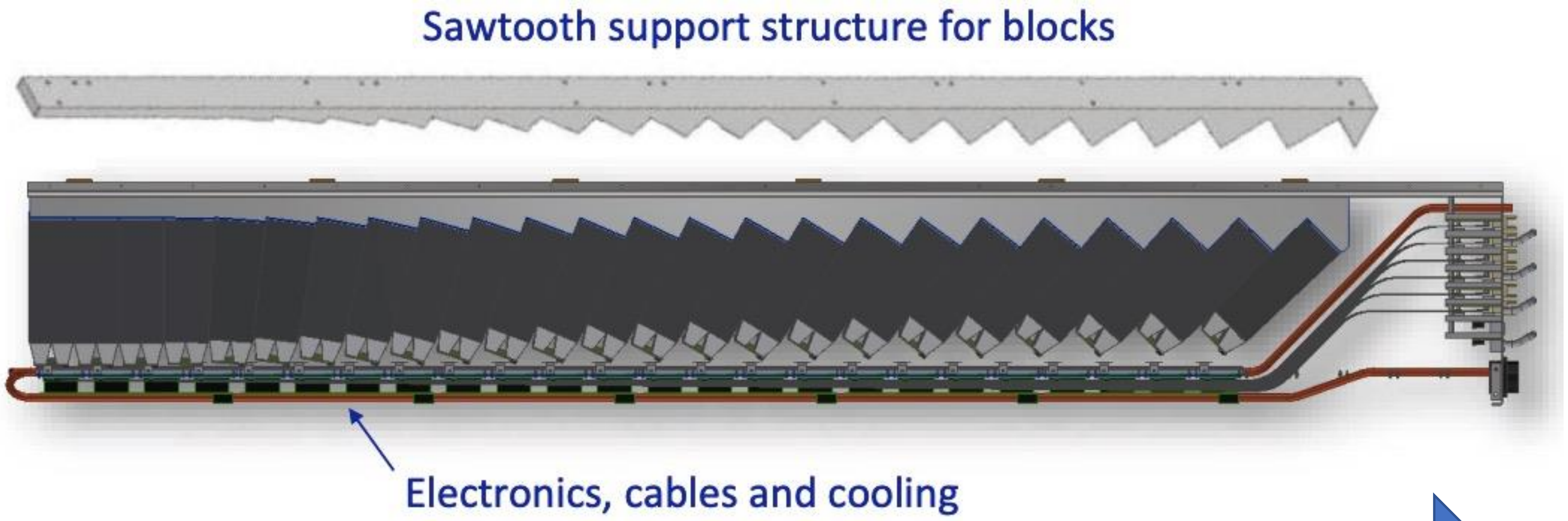


Electromagnetic Calorimetry - sPHENIX

- Tungsten powder, scintillator, embedded fibers
- Also 2D projective







Closer to collision → Further from collision

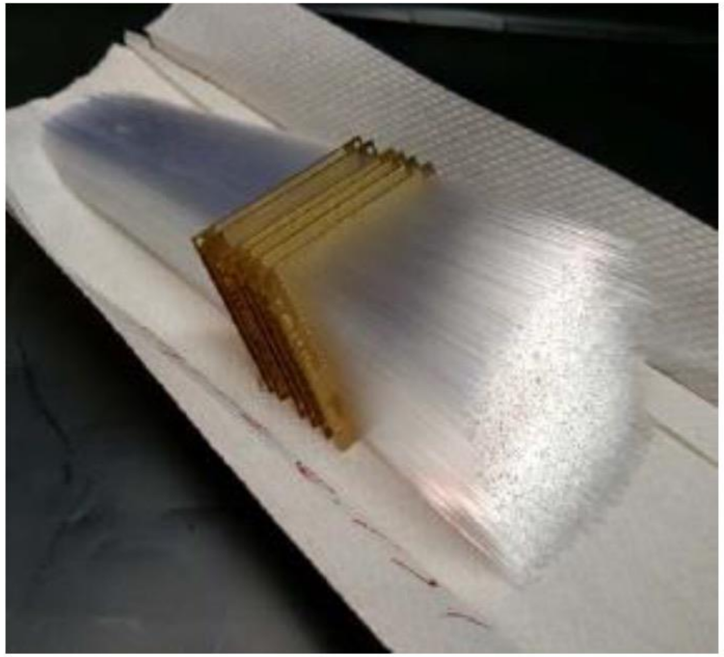
EMCal Blocks

Absorber

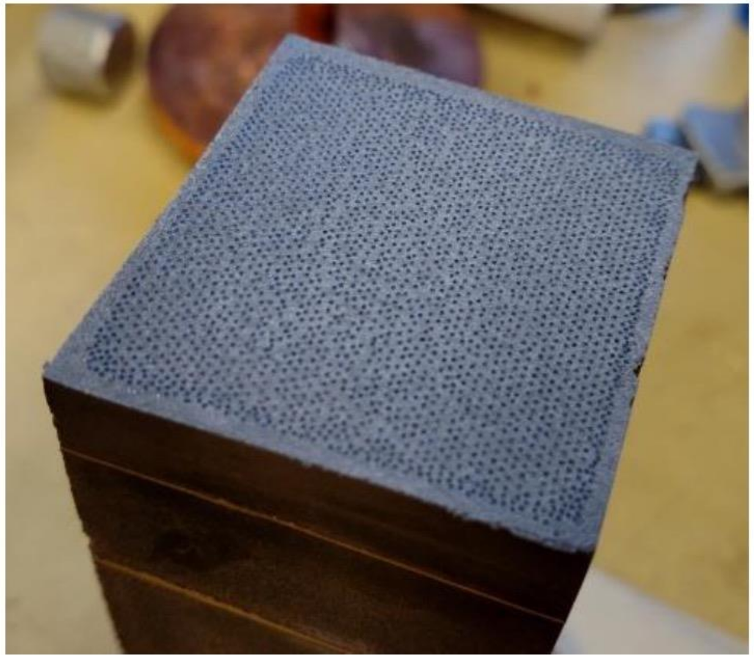
- Matrix of tungsten powder and epoxy with embedded scintillating fibers
  - Density ~ 9-10 g/cm3
  - X0~ 7 mm (18 X_0 total), R_M~ 2.3 cm

Scintillating fibers

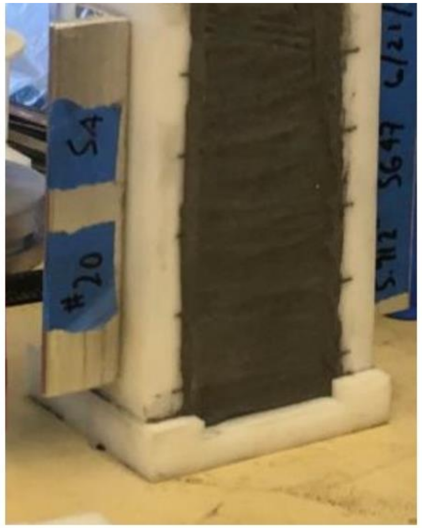
- Diameter: 0.47 mm, Spacing: 1 mm
- Sampling Fraction ~ 2 %



Fiber Assembly



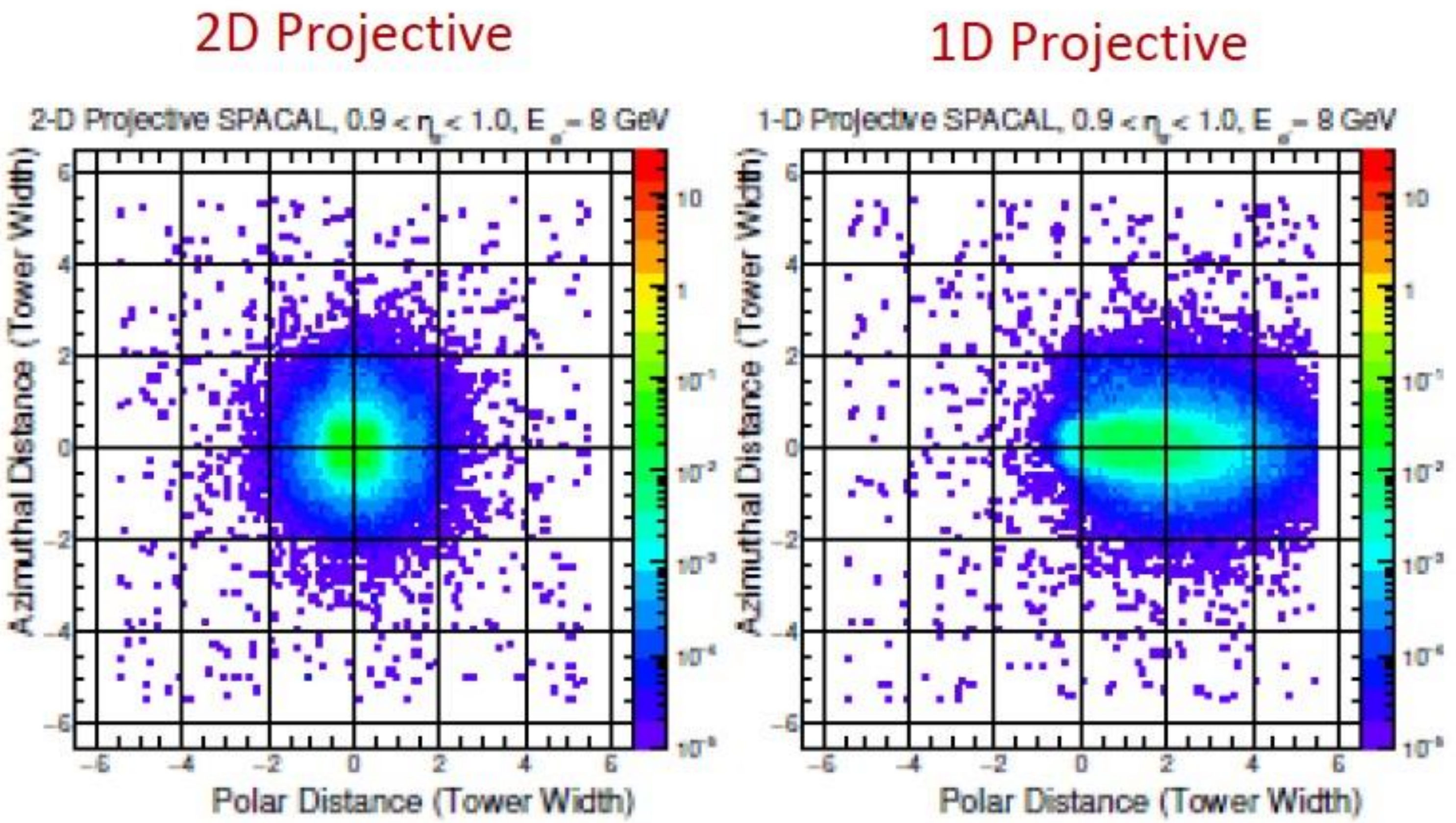
Fibers are tapered inward at readout end to improve light collection



Finished 2D projective block

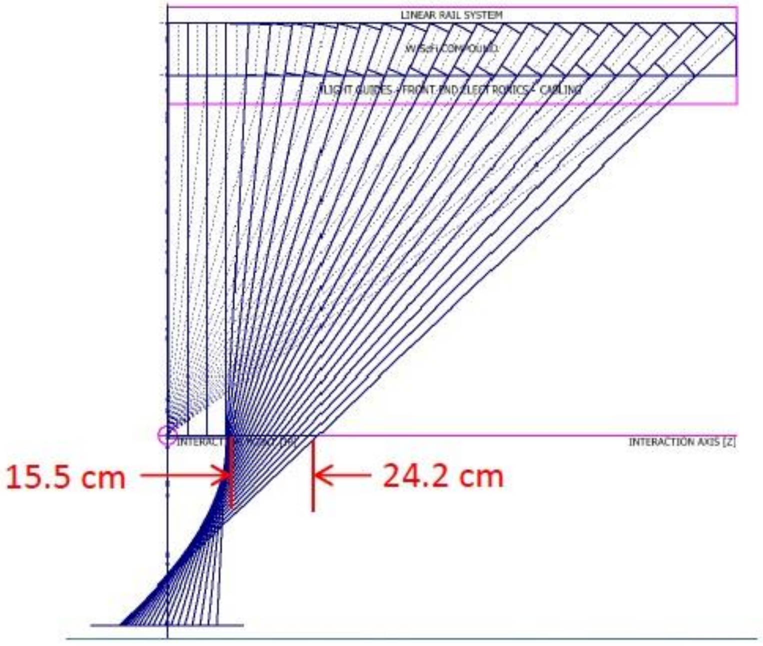
Electromagnetic Calorimetry - sPHENIX

- Why projectivity?
- Base case, fewer towers, less pedestal stochasticity
- Better case, concentrates traverse shower energy in localized region → more robust against UE
- Would also hazard a guess this improves efficacy of our shower shape variables

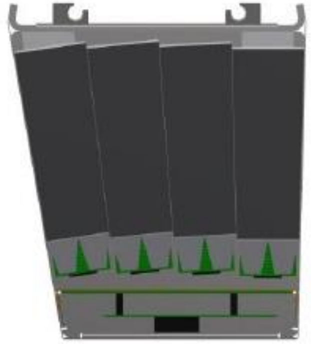


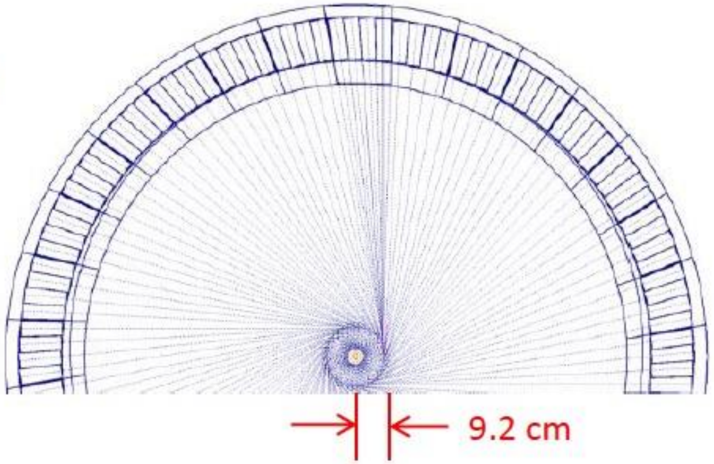
Electromagnetic Calorimetry - sPHENIX

- EMCal blocks are also tilted in azimuth as well
- The observant physicist might ask the reasonable question:
  - Wtf
- To prevent *channeling* whereby particles bypass shower production by passing through the active material
  - The fibers



Blocks are tilted in η
Sectors are tilted in ϕ





"At first sight, the calibration of a calorimeter system may seem to be straightforward and trivial. **It is not**."

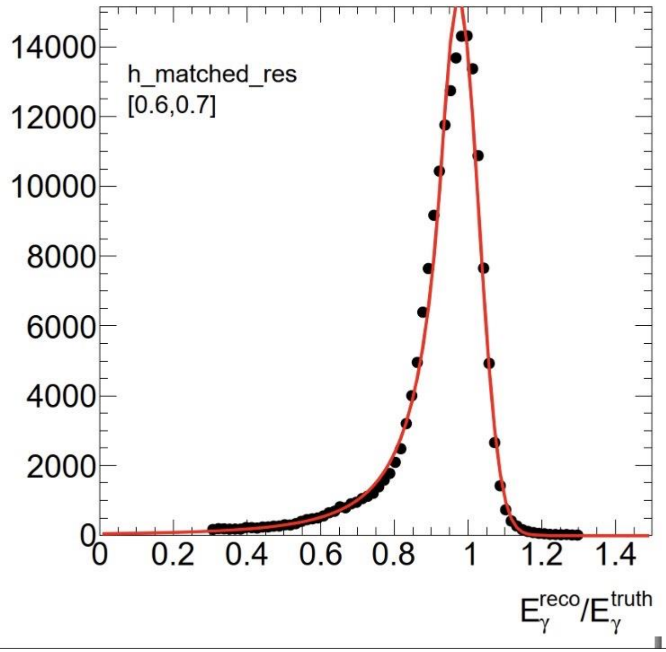
-Richard Wigmans

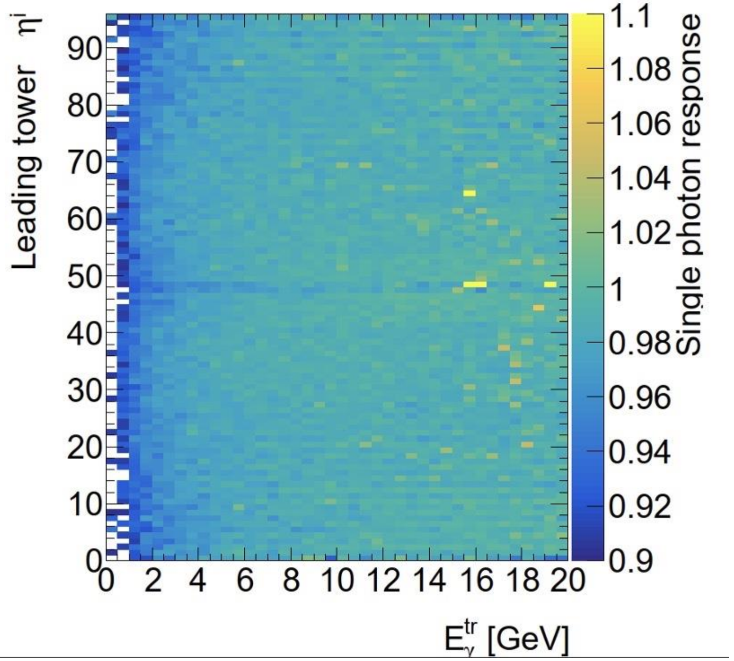
Calibrating the EMCal

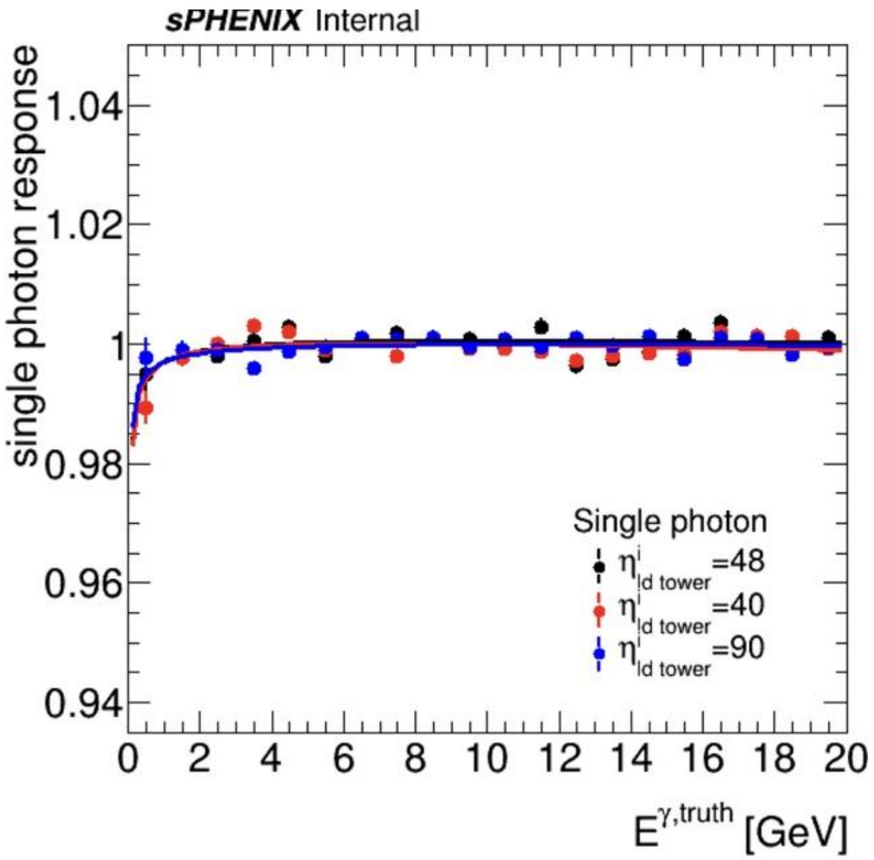
Actually Fairly Straightforward

EMCal Calibration Procedure

- I'm going to steal a lot of this from [Blair's slides from the CM]()
- Firstly, we set the G4 sPHENIX EMCal response to 1 with single photons

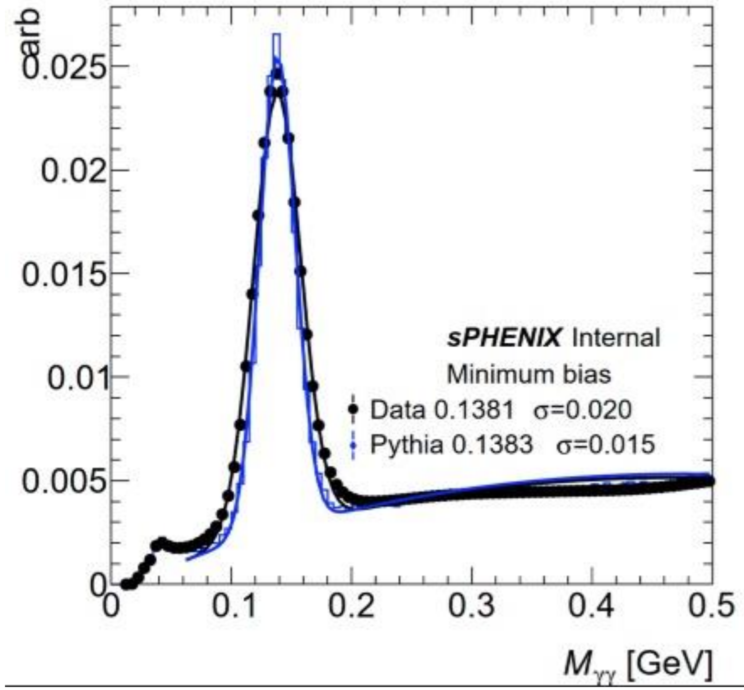


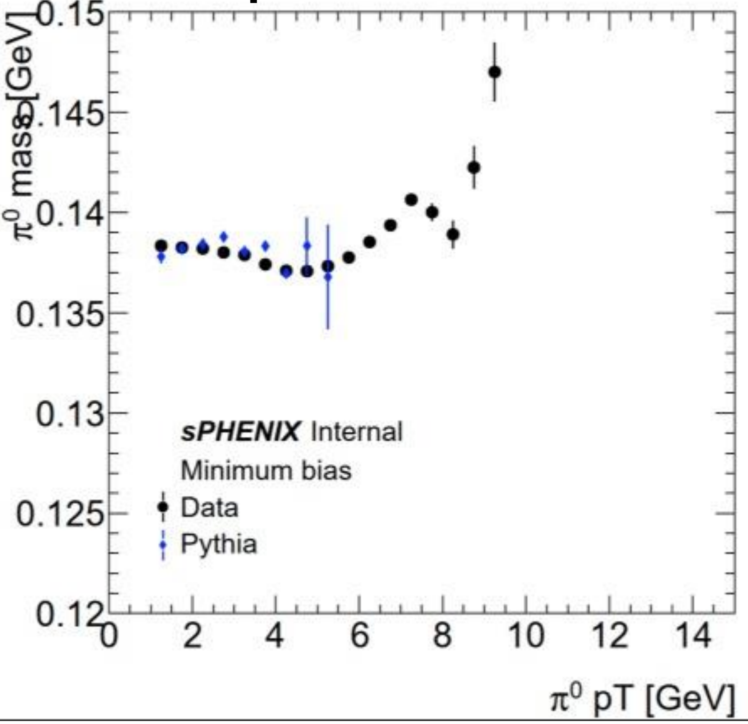


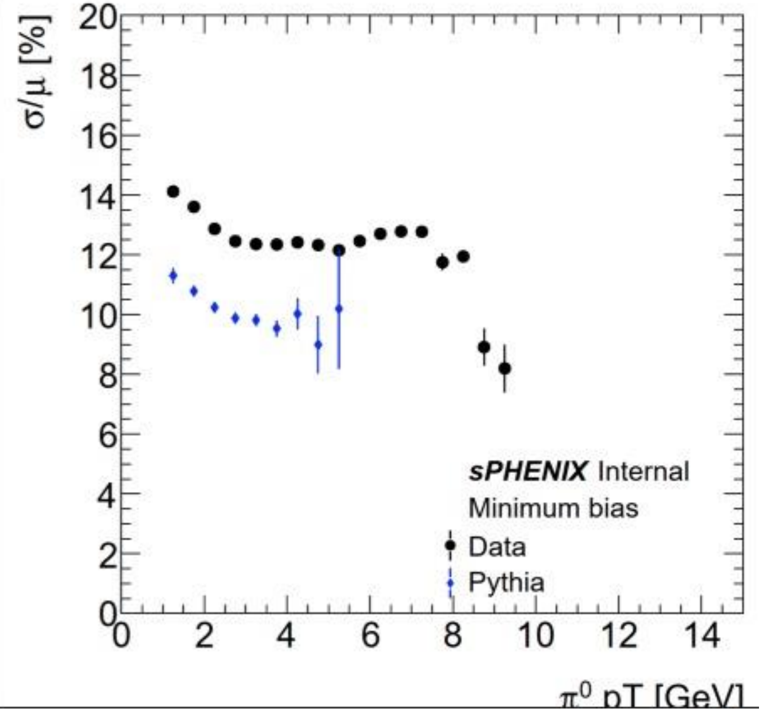


EMCal Calibration Procedure

- Next, we reconstruct π^0's from a "reasonable" looking spectrum
  - Largely PYTHIA
- We also try to match what we believe the real life detector's performance to be (e.g. mass peak width/ resolution).

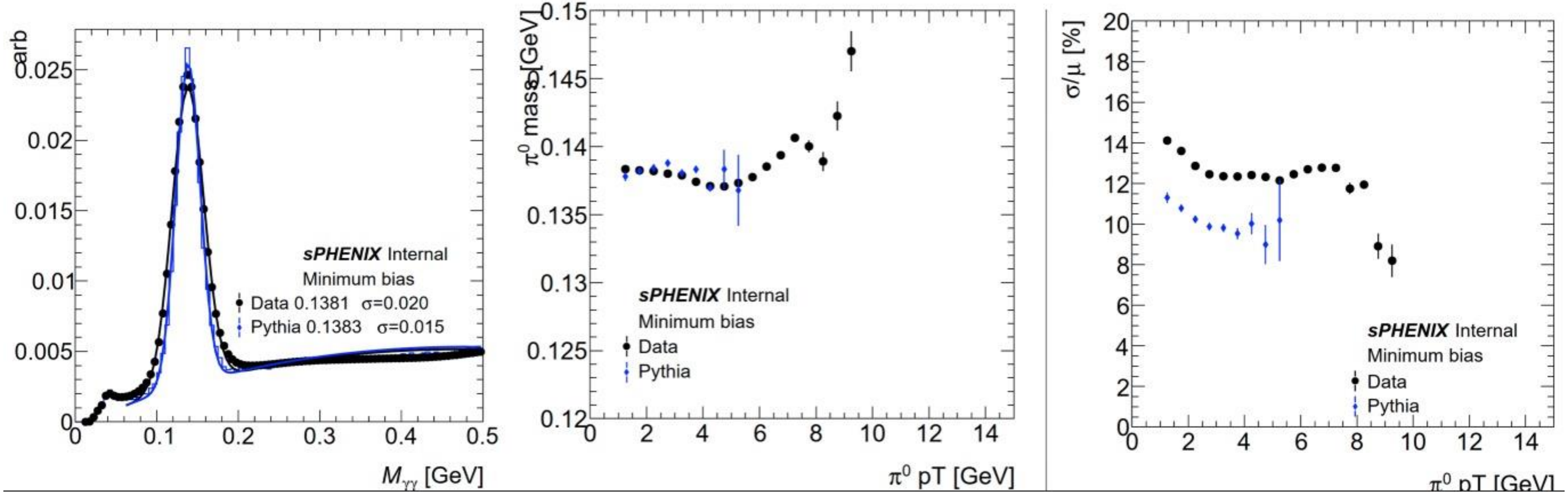






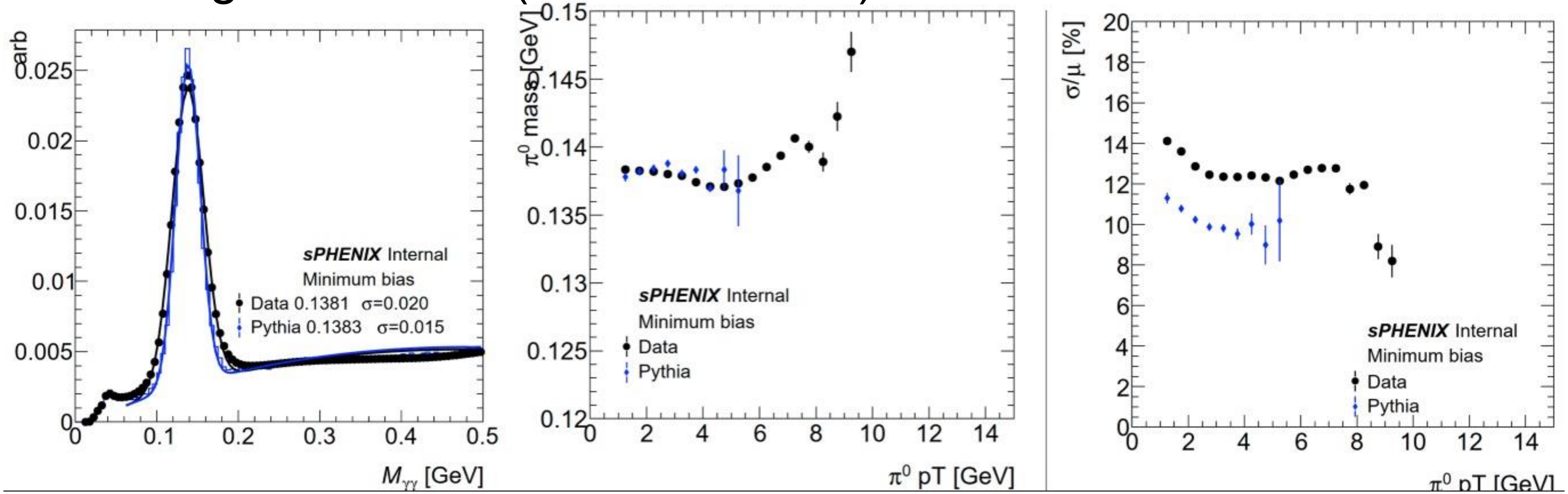
EMCal Calibration Procedure

- The points in blue here give us our “target mass” which we in turn tune the black points to (data)



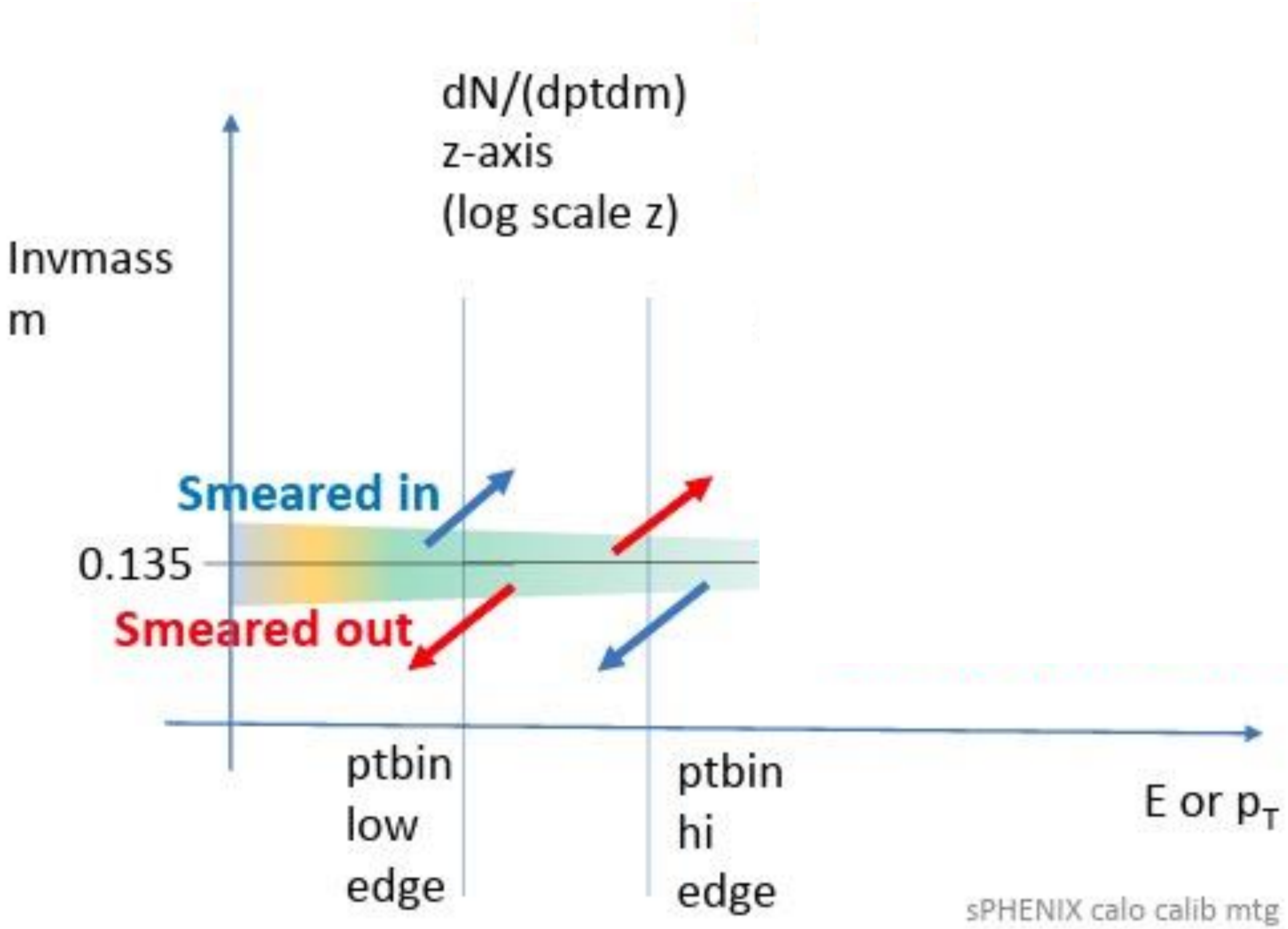
EMCal Calibration Procedure

- The points in blue here give us our “target mass” which we in turn tune the black points to (data)
- The astute among you might ask “why the heck do we do this instead of calibrating to 135MeV (the PDG mass)?



EMCal Calibration Procedure

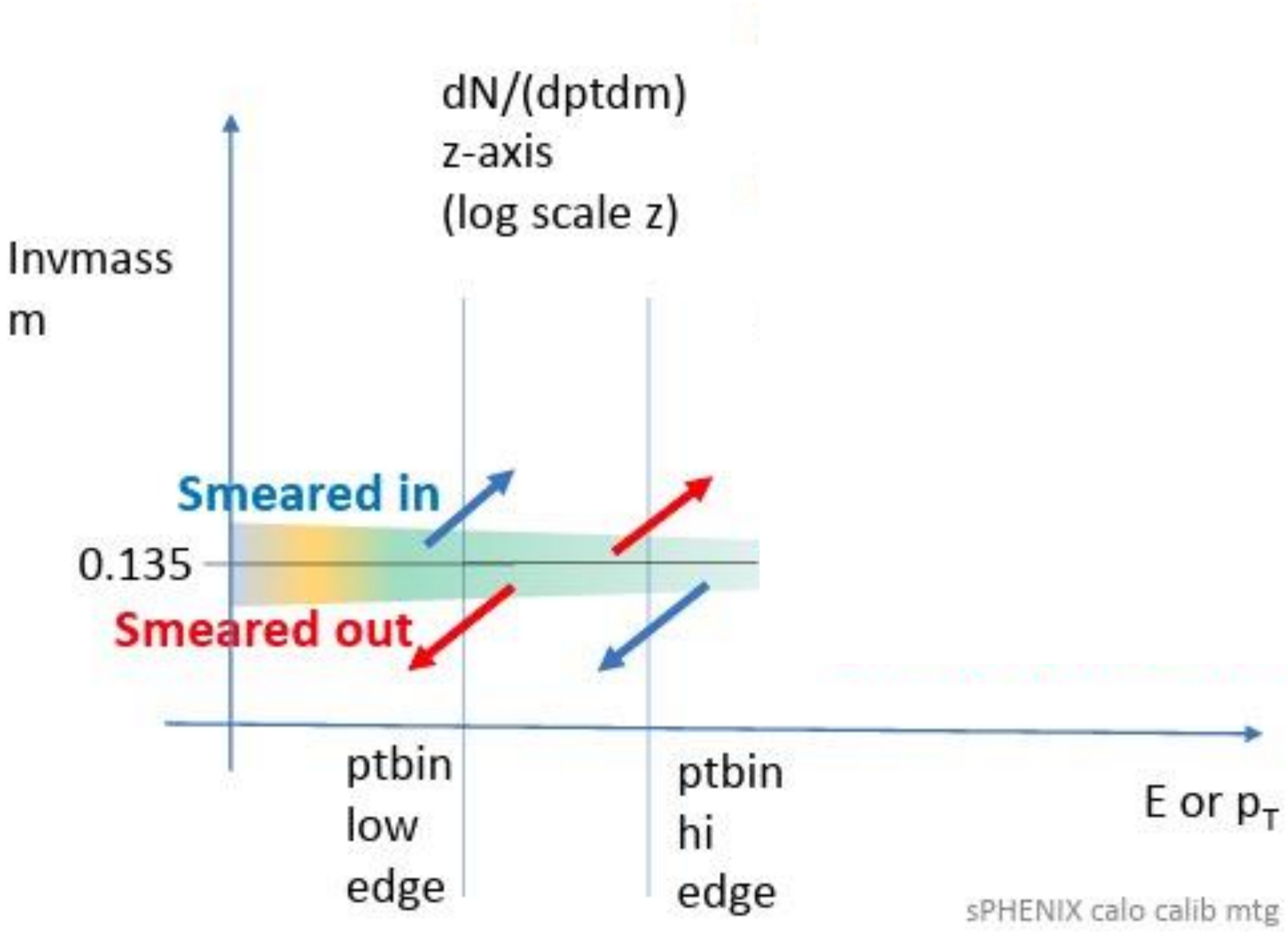
- It's really a **choice** but an appropriate one given that π^0 spectra are what we can measure in data
- The issue comes from two important aspects of measurements in general
  - The scale, μ
  - The resolution, σ/μ
- While we can set the scale perfectly, the resolution will shift the π^0 mass



Cartoon from [Justin Frantz (04/17/2023)]

EMCal Calibration Procedure

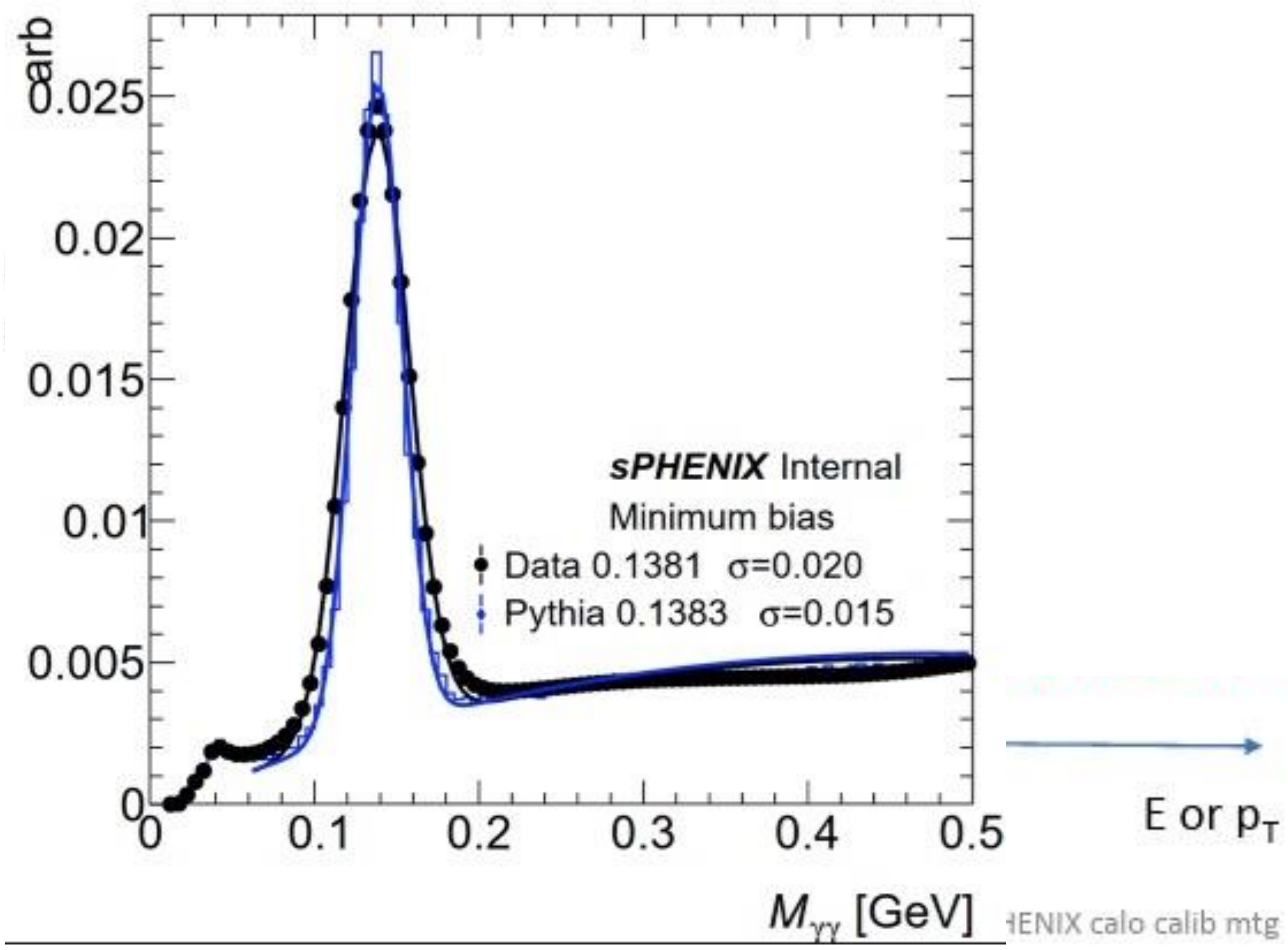
- Put simply, looking at the cartoon, the 135MeV region is full of upwards-smeared photons, not necessarily real $\pi^{0\prime}s$
- The region just north of it actually has a large contribution from real π^0's
- Note that you **can** calibrate to 135MeV, you just essentially eat a resolution-induced hit to the efficiency later one



Cartoon from [Justin Frantz (04/17/2023)]

EMCal Calibration Procedure

- Put simply, looking at the cartoon, the 135MeV region is full of upwards-smeared photons, not necessarily real $\pi^{0\prime}s$
- The region just north of it actually has a large contribution from real π^0's
- And indeed, you see this slight shift in the mass peak in sim and data



Cartoon from [Justin Frantz (04/17/2023)]

EMCal Calibration Procedure

- You've also probably heard the π^0 calibration be referred to as "iterative"
  - Because it is
- This is because the calibration factors are derived from calibrating the leading tower in each cluster
- For clusters with a large mount of energy in a single tower, this is fine
- For more diffuse clusters, obviously this introduces some cross-talk, and the iterative method handles this

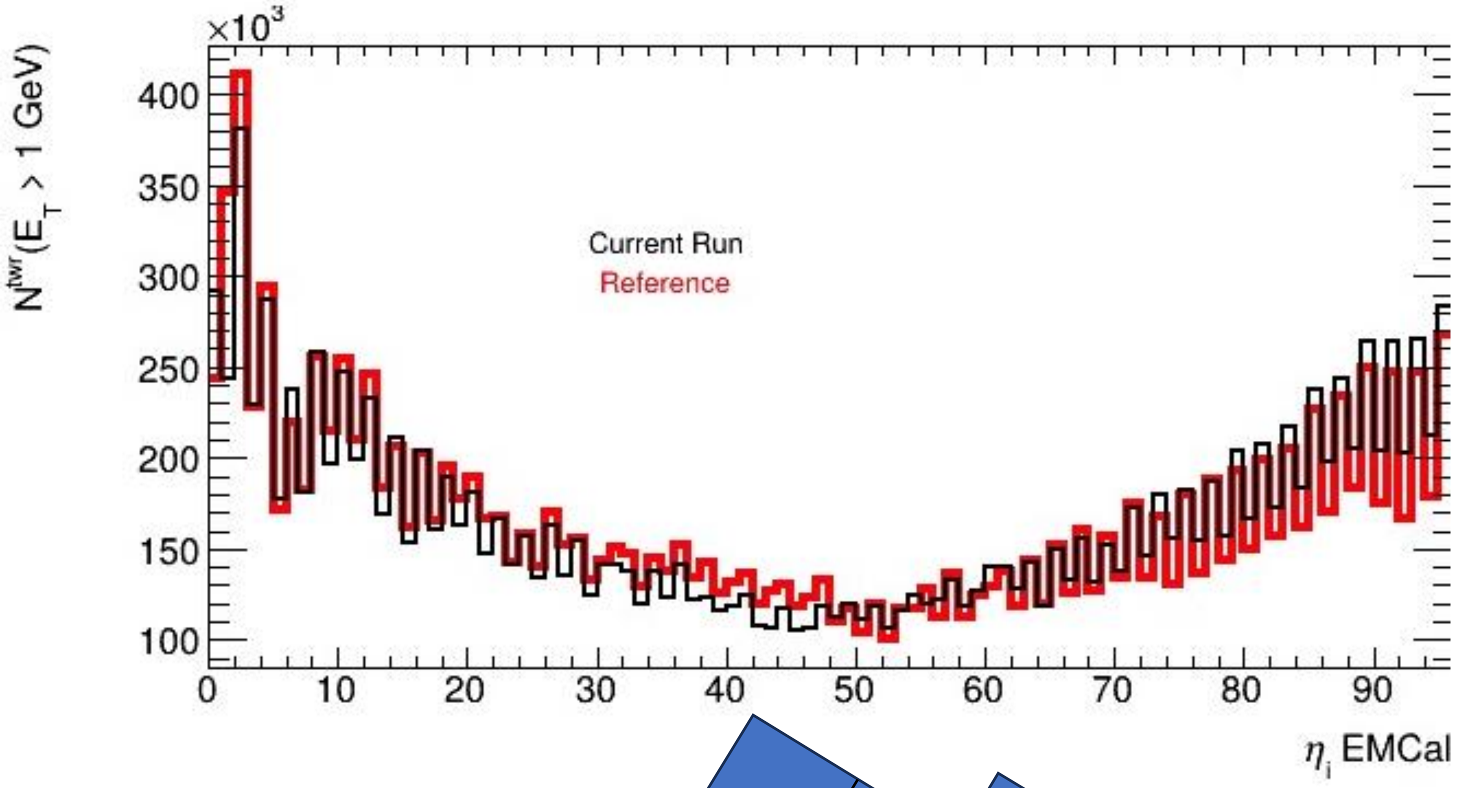
Wacky EMCal Junk

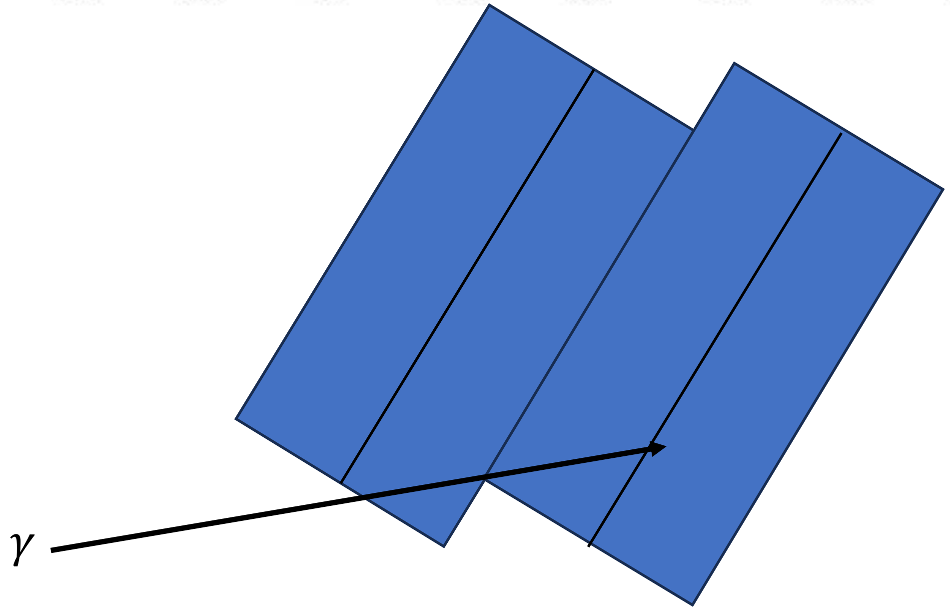
You: “Hey have you guys seen this befo-”

Yes, yes we have.

Tower Shadowing

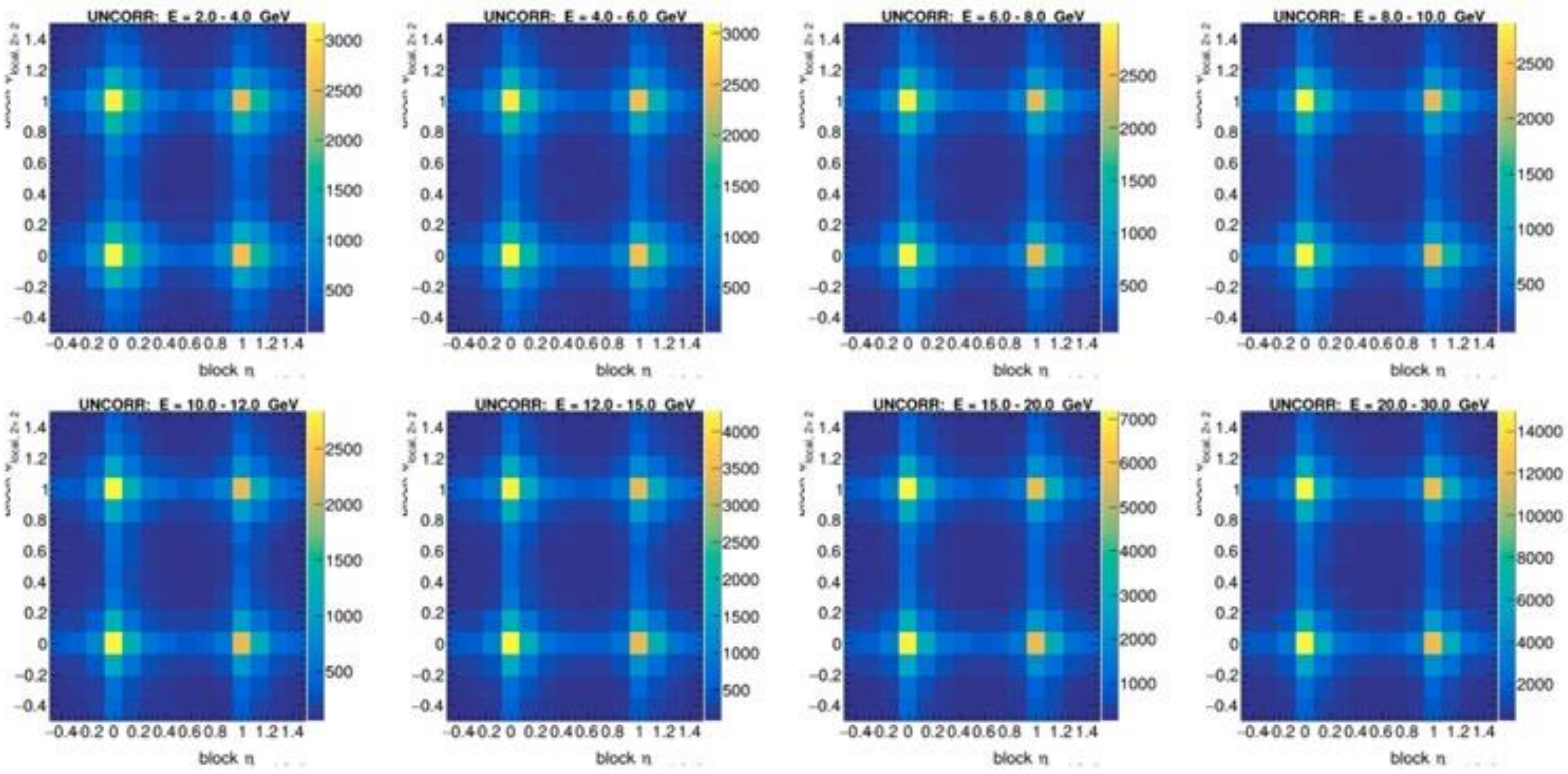
- If you look at the offline QA or even a basic analysis macro, you'll see this saw-tooth pattern to the tower energies
- This is "shadowing" induced by the block geometry, and is normal





Position Dependent Position

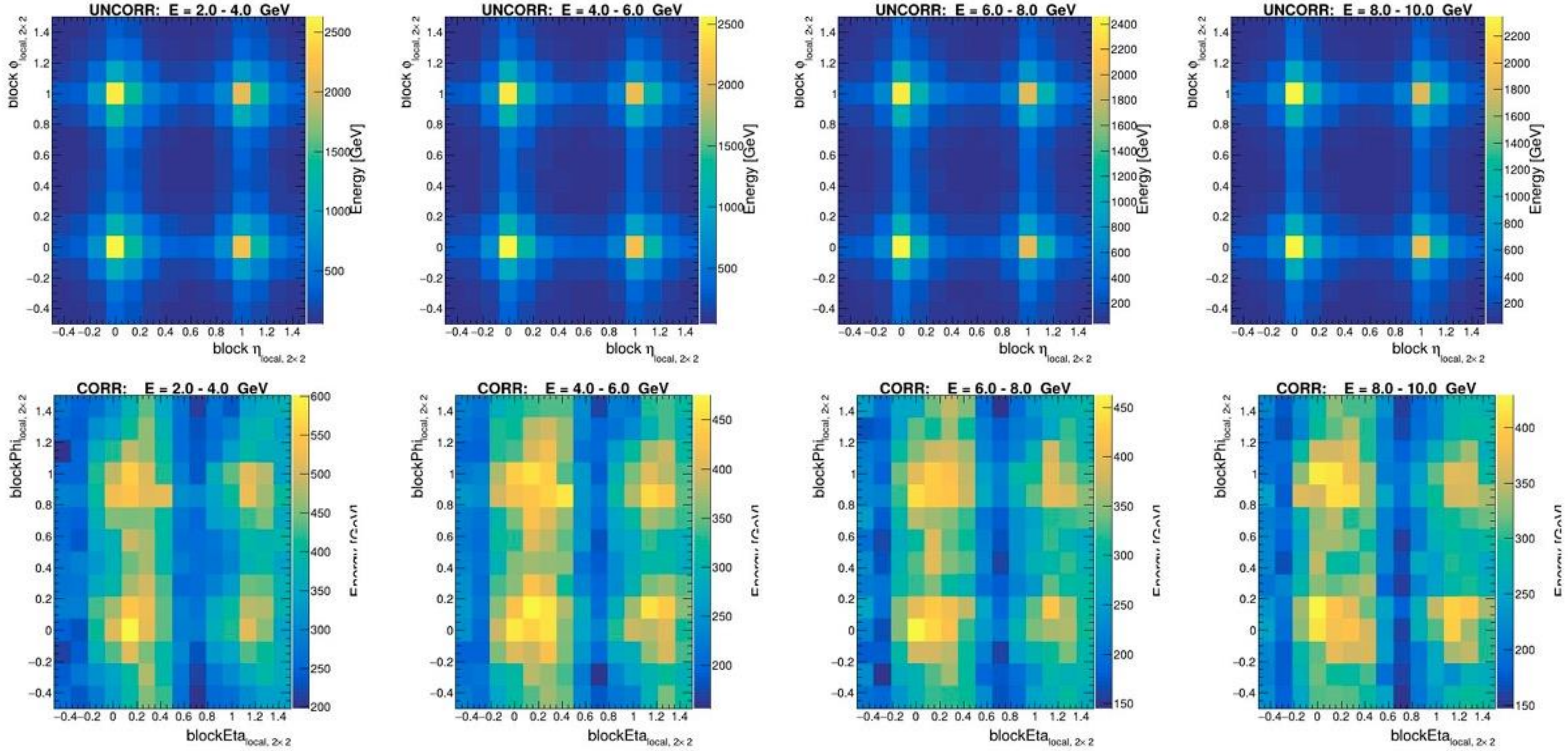
- If you took a very granular look at the cluster positions, you'd find they're basically always at the center of towers
- This is a bias in the reconstruction



Plots from [Justin Bennett (07/07//25)]

Position Dependent Response

- A little bit of intense mathematics allows you to correct this back to the unbiased positions



Tutorials

- A very cool guy™ made this tutorial
  - https://github.com/sPHENIX-Collaboration/tutorials/tree/master/CaloDataAnaRun24pp
  - Download: git@github.com:sPHENIX-Collaboration/tutorials.git
- It will allow you to access all the low-level calorimeter objects (towers) from the EMCal, HCal, and ZDC
- It also includes clusters
- Outputs all histograms, great for taking preliminary looks at low-level data
- Here's also a recording showing installation

Current Ways to Get Involved

- My opinion (directed at students *and* advisors): self-guided data exploration quickly plateaus with lower yields than you expect
- You need to talk to experts, attend meetings, and work on tasks
  - It's the difference between wandering around the gym picking up random dumbbells and having a real workout
- Attend the calorimeter calibration meetings, Mondays at 11:30ET!
- Most recent indico: https://indico.bnl.gov/event/31545/
- Mailing list(s):
  - sphenix-emcal-l@lists.bnl.gov
  - sphenix-hcal-l@lists.bnl.gov
  - sphenix-calibration-l@lists.bnl.gov

sPHENIX Calorimetry References

- 1st Test Beam: https://arxiv.org/abs/1704.01461
- 2nd Test Beam: https://arxiv.org/abs/2003.13685
- Technical Design Report: https://indico.bnl.gov/event/7081/attachments/25527/38284/sphenix_tdr_20190513.pdf

Parameter	Units	Value
Inner radius (envelope)	mm	900
Outer radius (envelope)	mm	1161
Length (envelope)	mm	$2 \times 1495 = 2990$
tower length (absorber)	mm	144
Number of towers in azimuth ($\Delta\phi$)		256
Number of towers in pseudorapidity ($\Delta\eta$)		$2 \times 48 = 96$
Number of electronic channels (towers)		$256 \times 96 = 24576$
Number of SiPMs per tower		4
Number of towers per sector		384
Number of sectors		$2 \times 32 = 64$
Sector weight (estimated)	kg	326
Total weight (estimated)	kg	20890
Average sampling fraction		2.3%

Table 3.2: Key parameters of the EMCal modules and sectors

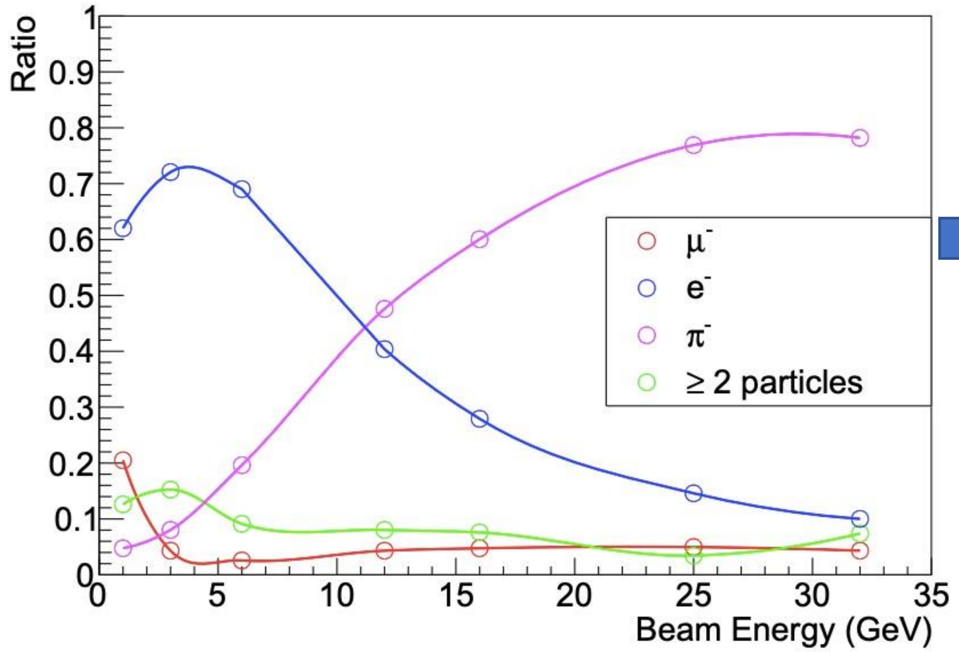
Back Up

Test Beams

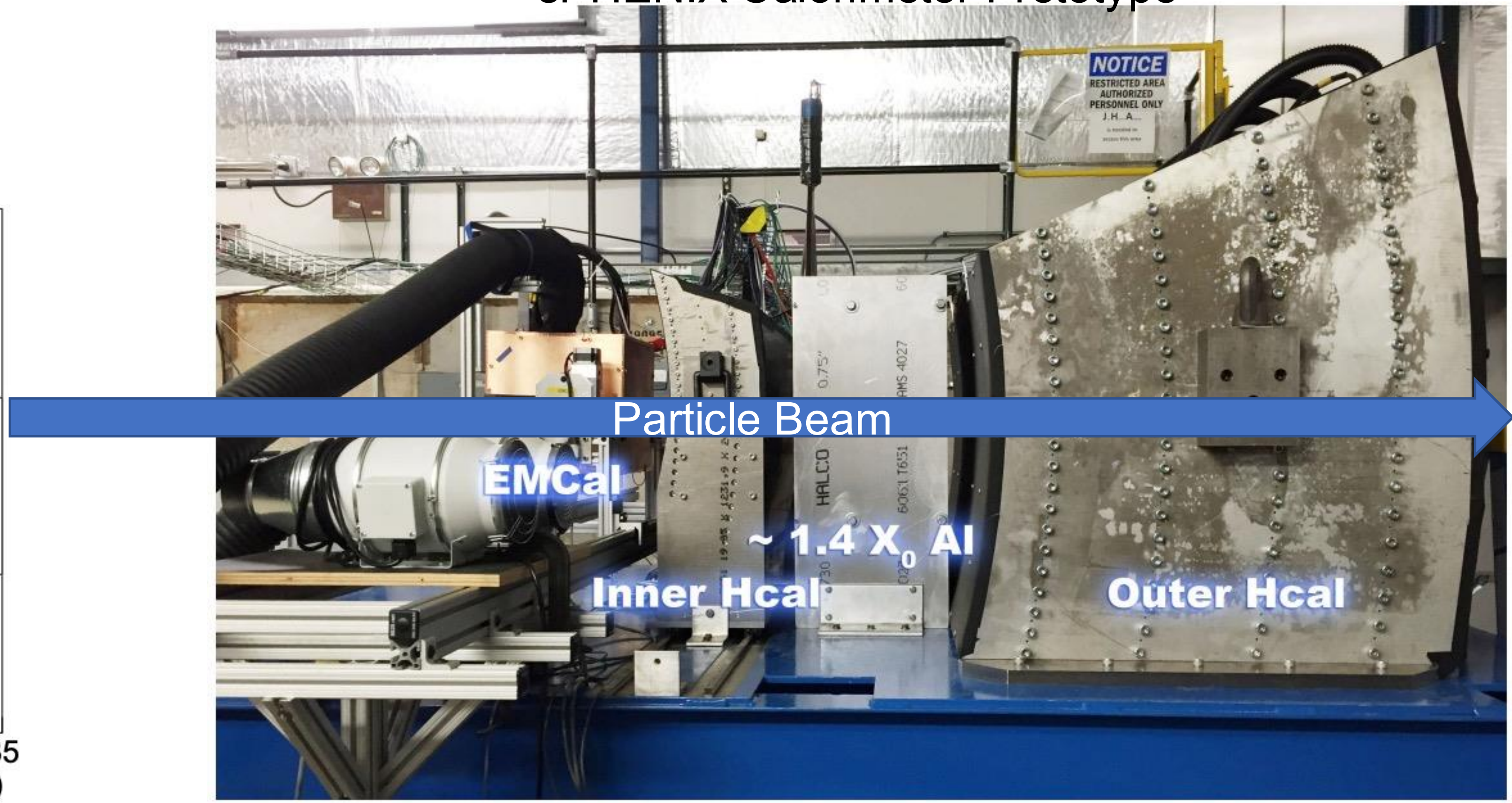
Quantifying Calorimeter Performance

- How do we tell how **good** or **not-so-good? Test Beams**

Fire a beam of particles of known **energy** and **composition** into detector prototype



sPHENIX Calorimeter Prototype



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Calorimeter Linearity

- **Linearity:** extent to which a detector's performance **changes** with **changing input energy**

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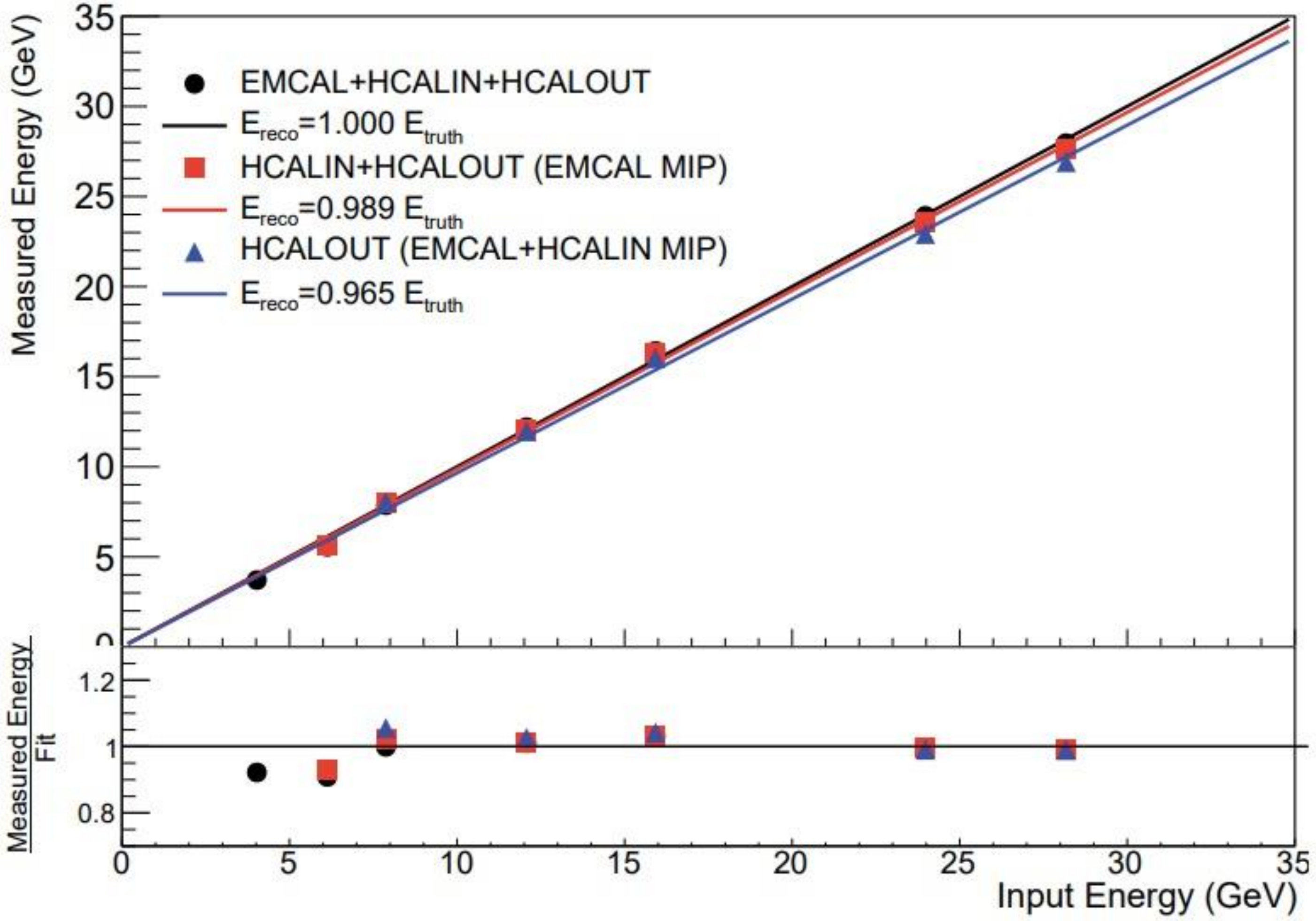

Calorimetric Resolution

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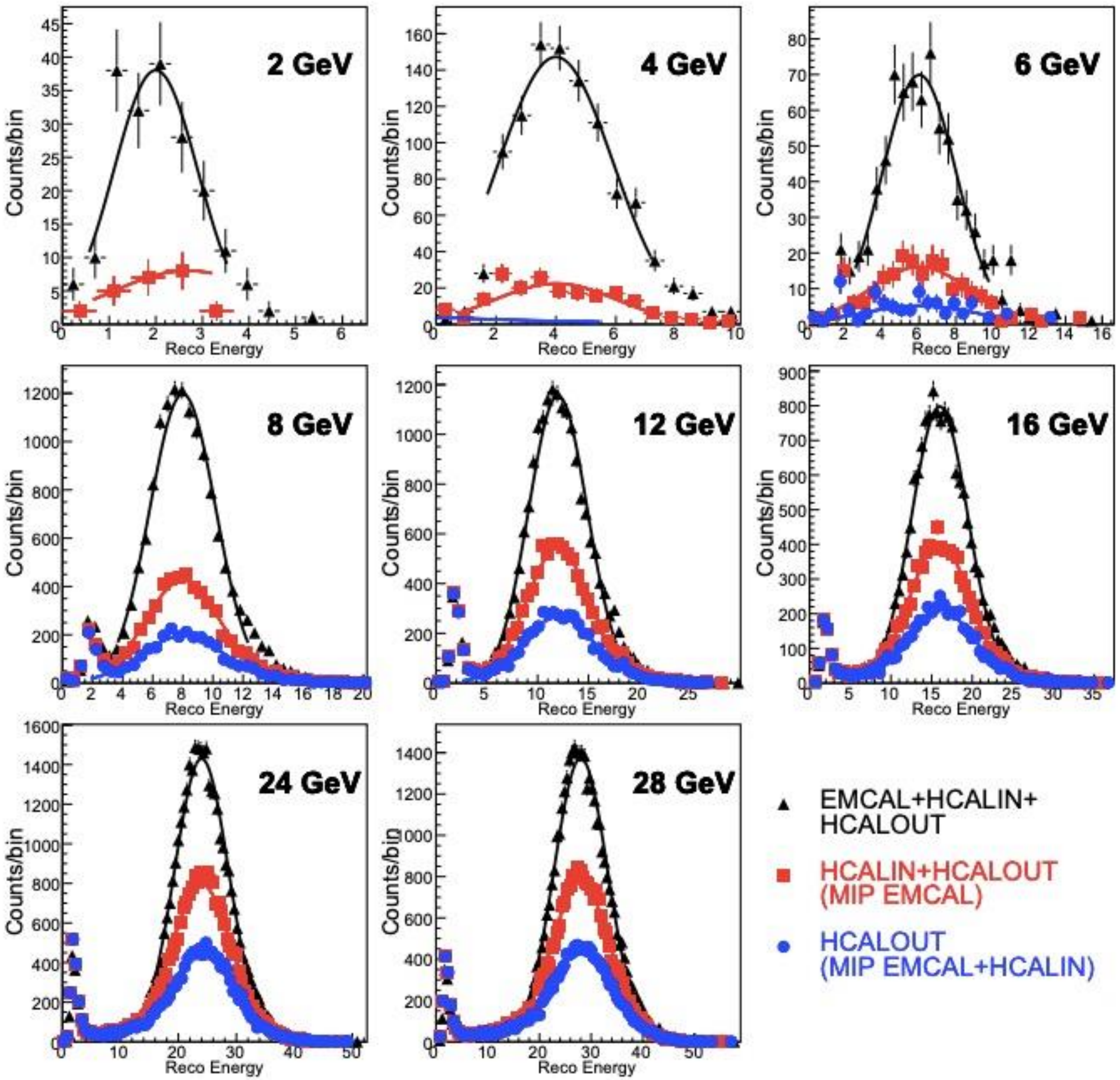
- **Resolution** encodes how **consistently** your detector performs
- Expressed as the relative error, $\frac{\sigma_E}{\langle E \rangle}$, of the distributions at right vs. input energy
- These plots are for **hadrons**



Calorimetric Resolution

- **Resolution** encodes how **consistently** your detector performs
- $\sigma_E/\langle E\rangle = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus$ c
- Stochastic term: fluctuations
  - This is typically quoted as the “energy resolution”, the physics performance parameter
  - sPHENIX requirement:
    - $a < 100\%$ for HCal
    - $a < 15\%$ for EMCal

For **hadrons**

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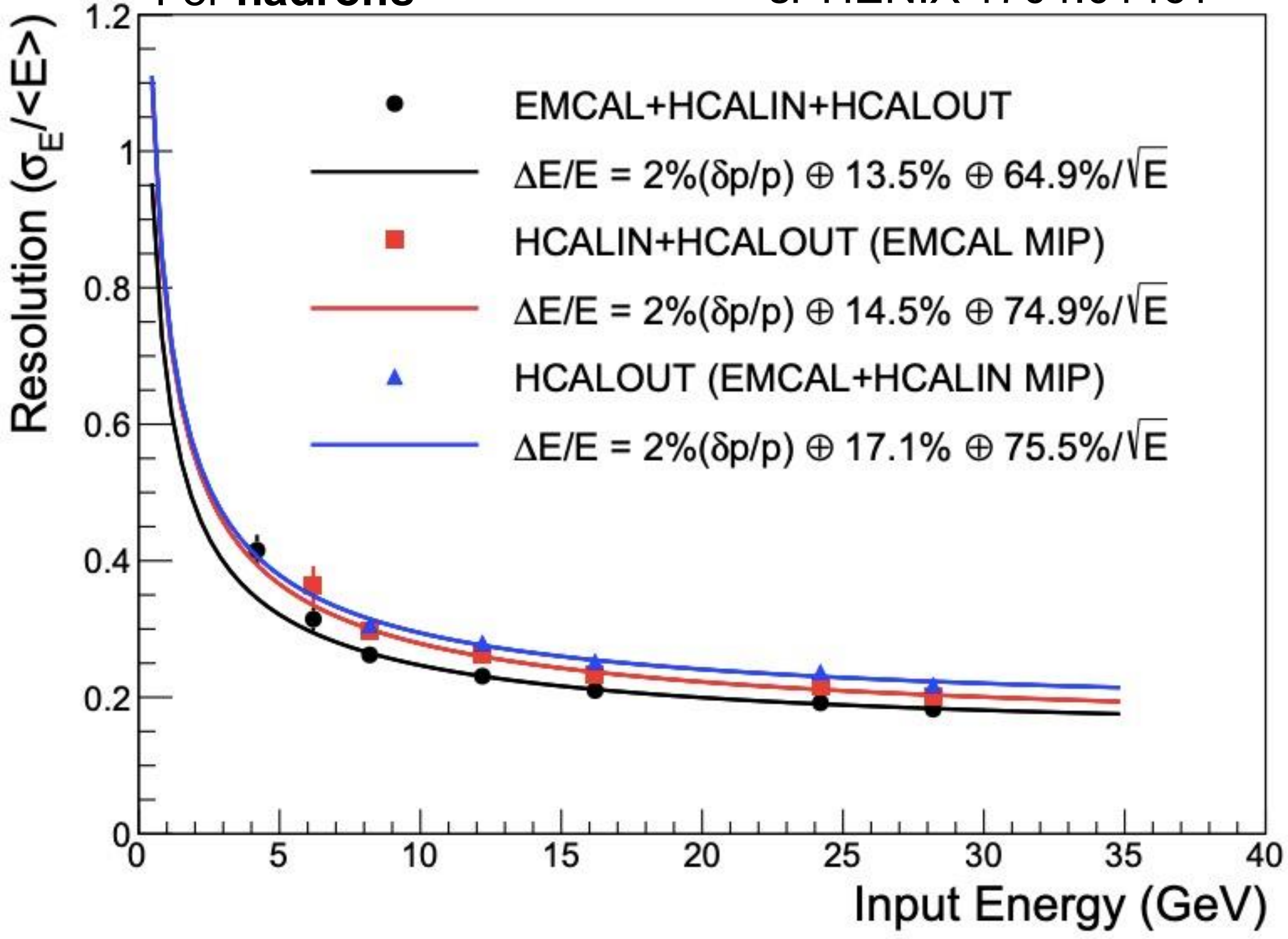
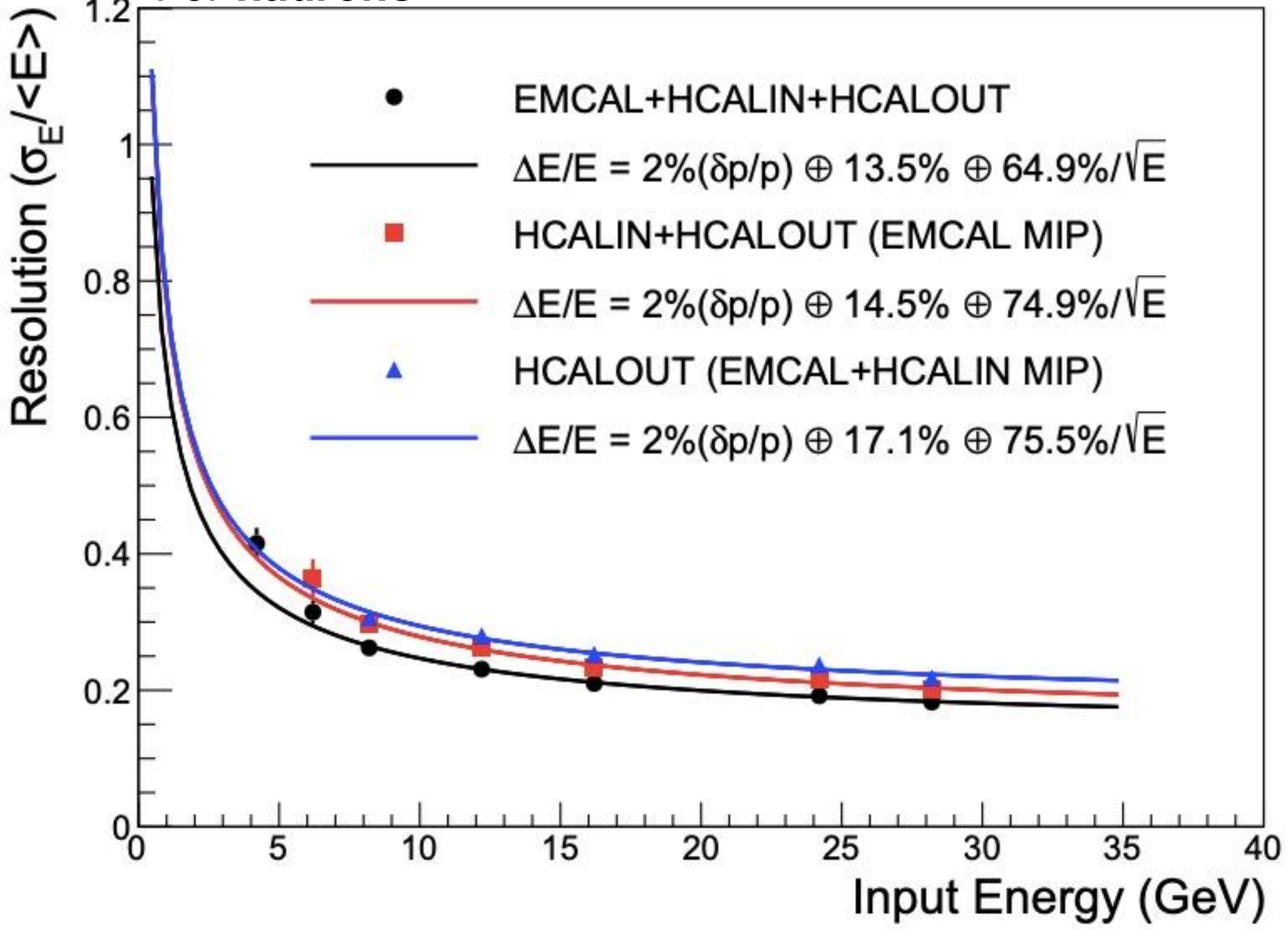


Calorimetric Resolution

- **Resolution** encodes how **consistently** your detector performs
- $\sigma_E/\langle E\rangle = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus$ c
- Noise term: electronic noise

For **hadrons**



Calorimetric Resolution

- **Resolution** encodes how **consistently** your detector performs
- $\sigma_E/\langle E\rangle = \frac{a}{\sqrt{E}} \oplus \frac{b}{E} \oplus$ c
- Constant term: everything else

For **hadrons**

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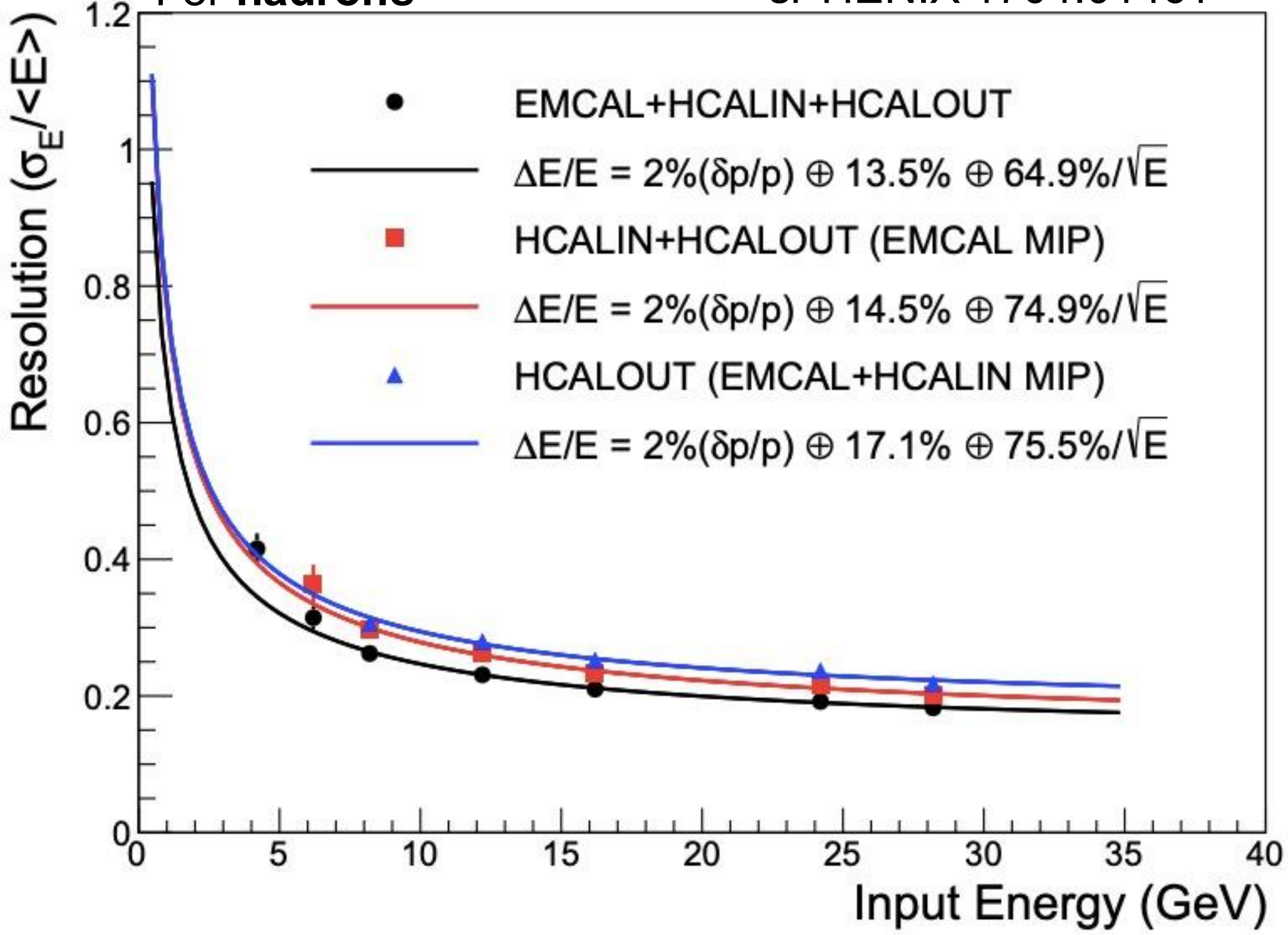

More Wigman Knowledge

Big Calorimetry Guy

Calibrating Segmented Calorimeters

"The correct way to intercalibrate the different sections of a longitudinally segmented, non-compensating calorimeter system is by using the same particles for all individual sections. If these particles develop showers, then they can only be used to calibrate sections in which these showers are completely contained."

-Richard Wigmans

[Calorimetry: Energy Measurement in Particle Physics]